In [49]:
```
# ============================================================
# STEP 1 — NLP PROJECT SETUP AND DATASET VERIFICATION
# ============================================================

import os
import sys
import pandas as pd
import numpy as np
from datetime import datetime

# Define dataset path
file_path = r"C:\Users\admin\OneDrive\Desktop\Activity-2\Hotel_Reviews.csv"

# Define NLP output folder
folder = os.path.join("output", "nlp_results")
os.makedirs(folder, exist_ok=True)

# Collect project setup information
setup_summary = {
    "Project": "Hotel Booking Cancellation Prediction and Analysis",
    "Stage": "Activity 2 — NLP Guest Feedback Analysis",
    "Dataset_File": file_path,
    "Dataset_Exists": os.path.exists(file_path),
    "Python_Version": sys.version.split()[0],
    "Pandas_Version": pd.__version__,
    "NumPy_Version": np.__version__,
    "Output_Folder": os.path.abspath(folder),
    "Dataset_Size_MB": (
        round(os.path.getsize(file_path) / (1024 * 1024), 2)
        if os.path.exists(file_path)
        else "File not found"
    ),
    "Date": datetime.now().strftime("%Y-%m-%d %H:%M:%S")
}

setup_df = pd.DataFrame(
    list(setup_summary.items()),
    columns=["Item", "Result"]
)

# Save setup summary
output_file = os.path.join(
    folder, "step_01_nlp_project_setup.csv"
)

setup_df.to_csv(output_file, index=False)

# Display results
print("=" * 70)
print("STEP 1 — NLP PROJECT SETUP AND DATASET VERIFICATION")
print("=" * 70)

print(setup_df.to_string(index=False))

print("\nSaved to:")
print(os.path.abspath(output_file))

if os.path.exists(file_path):
```

```
    print("\nSTEP 1 COMPLETED SUCCESSFULLY")
else:
    print("\nSTEP 1 NEEDS ATTENTION: Dataset file not found.")
    print("Ensure Hotel_Reviews.csv is in the same folder as the notebook.")
```

```
======================================================================
STEP 1 — NLP PROJECT SETUP AND DATASET VERIFICATION
======================================================================
           Item                                                    Result
        Project            Hotel Booking Cancellation Prediction and Analysis
          Stage                        Activity 2 — NLP Guest Feedback Analysis
   Dataset_File  C:\Users\admin\OneDrive\Desktop\Activity-2\Hotel_Reviews.csv
 Dataset_Exists                                                      True
 Python_Version                                                    3.14.6
 Pandas_Version                                                     3.0.5
  NumPy_Version                                                     2.5.2
  Output_Folder C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results
Dataset_Size_MB                                                    227.12
           Date                                          2026-10-23 19:51:40

Saved to:
C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_01_nlp_project
_setup.csv

STEP 1 COMPLETED SUCCESSFULLY
```

In [50]:

```
# ============================================================
# STEP 2 — DATASET LOADING AND INITIAL INSPECTION
# ============================================================

import os
import pandas as pd
from datetime import datetime

# Dataset path
file_path = "Hotel_Reviews.csv"

# NLP output folder
folder = os.path.join("output", "nlp_results")
os.makedirs(folder, exist_ok=True)

# Read only the first 5 rows for initial inspection
sample_df = pd.read_csv(
    file_path,
    nrows=5,
    low_memory=False
)

# Collect dataset information
inspection_summary = {
    "Dataset_File": os.path.basename(file_path),
    "Columns_Count": len(sample_df.columns),
    "Column_Names": ", ".join(sample_df.columns.astype(str)),
    "Sample_Rows_Read": len(sample_df),
    "Positive_Review_Column_Present": (
        "Positive_Review" in sample_df.columns
    ),
    "Negative_Review_Column_Present": (
        "Negative_Review" in sample_df.columns
    ),
```

```
    "Reviewer_Score_Column_Present": (
        "Reviewer_Score" in sample_df.columns
    ),
    "Date": datetime.now().strftime("%Y-%m-%d %H:%M:%S")
}

inspection_df = pd.DataFrame(
    list(inspection_summary.items()),
    columns=["Item", "Result"]
)

# Save inspection summary
output_file = os.path.join(
    folder, "step_02_dataset_initial_inspection.csv"
)

inspection_df.to_csv(output_file, index=False)

# Display results
print("=" * 70)
print("STEP 2 — DATASET LOADING AND INITIAL INSPECTION")
print("=" * 70)

print("\nDataset inspection summary:")
print(inspection_df.to_string(index=False))

print("\nFirst 5 rows of the dataset:")
display(sample_df)

print("\nSaved to:")
print(os.path.abspath(output_file))

print("\nSTEP 2 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 2 — DATASET LOADING AND INITIAL INSPECTION
======================================================================

Dataset inspection summary:
                         Item
Result
                 Dataset_File
Hotel_Reviews.csv
                Columns_Count
17
                 Column_Names Hotel_Address, Additional_Number_of_Scoring, Revie
w_Date, Average_Score, Hotel_Name, Reviewer_Nationality, Negative_Review, Review_
Total_Negative_Word_Counts, Total_Number_of_Reviews, Positive_Review, Review_Tota
l_Positive_Word_Counts, Total_Number_of_Reviews_Reviewer_Has_Given, Reviewer_Scor
e, Tags, days_since_review, lat, lng
             Sample_Rows_Read
5
Positive_Review_Column_Present
True
Negative_Review_Column_Present
True
 Reviewer_Score_Column_Present
True
                         Date
2026-10-23 19:51:40

First 5 rows of the dataset:
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Hotel_Address</th> <th>Additional_Number_of_Scoring</th> <th>Review_Date</th> <th>Average_Score</th> <th>Hotel_Name</th> <th>Reviewer_Nationality</th> <th>Negative_Review</th> <th>Review_Total_Negative_Word_Counts</th> <th>Total_Number_of_Reviews</th> <th>Positive_Review</th> <th>Review_Total_Positive_Word_Counts</th> <th>Total_Number_of_Reviews_Reviewer_Has_Given</th> <th>Reviewer_Score</th> <th>Tags</th> <th>days_since_review</th> <th>lat</th> <th>lng</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

s Gravesandestraat 55 Oost 1092 AA Amsterdam ...	194	8/3/2017	7.7	Hotel Arena	Russia	I am so angry that i made this post available...	397	1403	Only the park outside of the hotel was beauti...
s Gravesandestraat 55 Oost 1092 AA Amsterdam ...	194	8/3/2017	7.7	Hotel Arena	Ireland	No Negative	0	1403	No real complaints the hotel was great great ...
s Gravesandestraat 55 Oost 1092 AA Amsterdam ...	194	7/31/2017	7.7	Hotel Arena	Australia	Rooms are nice but for elderly a bit difficul...	42	1403	Location was good and staff were ok It is cut...
s Gravesandestraat 55 Oost 1092 AA Amsterdam ...	194	7/31/2017	7.7	Hotel Arena	United Kingdom	My room was dirty and I was afraid to walk ba...	210	1403	Great location in nice surroundings the bar a...
s Gravesandestraat 55 Oost 1092 AA Amsterdam ...	194	7/24/2017	7.7	Hotel Arena	New Zealand	You When I booked with your company on line y...	140	1403	Amazing location and building Romantic setting

```
Saved to:
C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_02_dataset_initial_inspection.csv

STEP 2 COMPLETED SUCCESSFULLY
```

In [51]:
```
# ============================================================
# STEP 3 — DATASET SCHEMA AND COLUMN VALIDATION
```

```
# ============================================================

import os
import pandas as pd
from datetime import datetime

file_path = "Hotel_Reviews.csv"
folder = os.path.join("output", "nlp_results")
os.makedirs(folder, exist_ok=True)

sample_df = pd.read_csv(file_path, nrows=5, low_memory=False)

required_columns = [
    "Positive_Review",
    "Negative_Review",
    "Reviewer_Score"
]

validation_rows = []

for column in sample_df.columns:
    validation_rows.append({
        "Column": column,
        "Data_Type": str(sample_df[column].dtype),
        "Required_For_NLP": column in required_columns,
        "Present": True
    })

for column in required_columns:
    if column not in sample_df.columns:
        validation_rows.append({
            "Column": column,
            "Data_Type": "Missing",
            "Required_For_NLP": True,
            "Present": False
        })

schema_df = pd.DataFrame(validation_rows)
output_file = os.path.join(folder, "step_03_schema_validation.csv")
schema_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 3 — DATASET SCHEMA AND COLUMN VALIDATION")
print("=" * 70)
print(schema_df.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))

missing_required = [
    c for c in required_columns if c not in sample_df.columns
]

if missing_required:
    print("\nRequired columns missing:", missing_required)
    print("STOP: Check the dataset column names before continuing.")
else:
    print("\nAll required NLP columns are present.")
    print("STEP 3 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 3 — DATASET SCHEMA AND COLUMN VALIDATION
======================================================================
                                     Column Data_Type  Required_For_NLP  Present
                              Hotel_Address       str             False     True
                Additional_Number_of_Scoring     int64             False     True
                                Review_Date       str             False     True
                              Average_Score   float64             False     True
                                Hotel_Name       str             False     True
                       Reviewer_Nationality       str             False     True
                             Negative_Review       str              True     True
          Review_Total_Negative_Word_Counts     int64             False     True
                    Total_Number_of_Reviews     int64             False     True
                             Positive_Review       str              True     True
          Review_Total_Positive_Word_Counts     int64             False     True
Total_Number_of_Reviews_Reviewer_Has_Given     int64             False     True
                              Reviewer_Score   float64              True     True
                                        Tags       str             False     True
                           days_since_review       str             False     True
                                         lat   float64             False     True
                                         lng   float64             False     True

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_03_s
chema_validation.csv

All required NLP columns are present.
STEP 3 COMPLETED SUCCESSFULLY
```

In [52]:

```
# ============================================================
# STEP 4 — MISSING VALUES AND DUPLICATE INSPECTION
# ============================================================

import os
import pandas as pd
from datetime import datetime

file_path = "Hotel_Reviews.csv"
folder = os.path.join("output", "nlp_results")
os.makedirs(folder, exist_ok=True)

missing_counts = {}
duplicate_count = 0
rows_scanned = 0

for chunk in pd.read_csv(file_path, chunksize=50000, low_memory=False):
    rows_scanned += len(chunk)

    for column in chunk.columns:
        missing_counts[column] = (
            missing_counts.get(column, 0) + chunk[column].isna().sum()
        )

    duplicate_count += int(chunk.duplicated().sum())

quality_df = pd.DataFrame([
    {
        "Column": column,
        "Missing_Count": int(count),
        "Missing_Percentage": round(count / rows_scanned * 100, 2)
    }
```

```
    for column, count in missing_counts.items()
])

output_file = os.path.join(folder, "step_04_data_quality_report.csv")
quality_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 4 — MISSING VALUES AND DUPLICATE INSPECTION")
print("=" * 70)
print("Rows scanned:", rows_scanned)
print("Duplicate rows within individual chunks:", duplicate_count)
print("\nMissing value report:")
print(quality_df.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 4 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 4 — MISSING VALUES AND DUPLICATE INSPECTION
======================================================================
Rows scanned: 515738
Duplicate rows within individual chunks: 526

Missing value report:
                                  Column  Missing_Count  Missing_Percentage
                           Hotel_Address              0                0.00
          Additional_Number_of_Scoring              0                0.00
                             Review_Date              0                0.00
                           Average_Score              0                0.00
                              Hotel_Name              0                0.00
                     Reviewer_Nationality              0                0.00
                            Negative_Review              0                0.00
      Review_Total_Negative_Word_Counts              0                0.00
                Total_Number_of_Reviews              0                0.00
                         Positive_Review              0                0.00
      Review_Total_Positive_Word_Counts              0                0.00
Total_Number_of_Reviews_Reviewer_Has_Given              0                0.00
                           Reviewer_Score              0                0.00
                                     Tags              0                0.00
                          days_since_review              0                0.00
                                      lat           3268                0.63
                                      lng           3268                0.63

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_04_d
ata_quality_report.csv

STEP 4 COMPLETED SUCCESSFULLY
```

In [53]:

```
# ============================================================
# STEP 5 — CREATE REPRODUCIBLE NLP SAMPLE
# ============================================================

import os
import pandas as pd
from datetime import datetime

file_path = "Hotel_Reviews.csv"
folder = os.path.join("output", "nlp_results")
os.makedirs(folder, exist_ok=True)

sampled_chunks = []
```

```
for chunk_number, chunk in enumerate(
    pd.read_csv(file_path, chunksize=50000, low_memory=False)
):
    if len(chunk) > 2000:
        chunk_sample = chunk.sample(n=2000, random_state=42)
    else:
        chunk_sample = chunk

    sampled_chunks.append(chunk_sample)

reviews_df = pd.concat(sampled_chunks, ignore_index=True)

if len(reviews_df) > 20000:
    reviews_df = reviews_df.sample(
        n=20000, random_state=42
    ).reset_index(drop=True)

sample_path = os.path.join(folder, "step_05_nlp_sample.csv")
reviews_df.to_csv(sample_path, index=False)

print("=" * 70)
print("STEP 5 — CREATE REPRODUCIBLE NLP SAMPLE")
print("=" * 70)
print("Sample rows:", len(reviews_df))
print("Sample columns:", len(reviews_df.columns))
print("Sample saved to:", os.path.abspath(sample_path))
print("\nSTEP 5 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 5 — CREATE REPRODUCIBLE NLP SAMPLE
======================================================================
Sample rows: 20000
Sample columns: 17
Sample saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\st
ep_05_nlp_sample.csv

STEP 5 COMPLETED SUCCESSFULLY
```

In [54]:

```
# ============================================================
# STEP 6 — NLP SAMPLE VALIDATION
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
sample_path = os.path.join(folder, "step_05_nlp_sample.csv")

reviews_df = pd.read_csv(sample_path, low_memory=False)

required_columns = [
    "Positive_Review",
    "Negative_Review",
    "Reviewer_Score"
]

validation = {
    "Sample_File_Exists": os.path.exists(sample_path),
```

```
    "Sample_Rows": len(reviews_df),
    "Sample_Columns": len(reviews_df.columns),
    "Required_Columns_Present": all(
        column in reviews_df.columns for column in required_columns
    ),
    "Duplicate_Rows_In_Sample": int(reviews_df.duplicated().sum()),
    "Date": datetime.now().strftime("%Y-%m-%d %H:%M:%S")
}

validation_df = pd.DataFrame(
    list(validation.items()), columns=["Item", "Result"]
)

output_file = os.path.join(folder, "step_06_sample_validation.csv")
validation_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 6 — NLP SAMPLE VALIDATION")
print("=" * 70)
print(validation_df.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))

if validation["Required_Columns_Present"]:
    print("\nSTEP 6 COMPLETED SUCCESSFULLY")
else:
    print("\nSTOP: Required columns are missing.")
```

```
======================================================================
STEP 6 — NLP SAMPLE VALIDATION
======================================================================
                    Item              Result
     Sample_File_Exists                True
            Sample_Rows               20000
          Sample_Columns                  17
Required_Columns_Present                True
Duplicate_Rows_In_Sample                   1
                    Date 2026-10-23 19:51:53

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_06_s
ample_validation.csv

STEP 6 COMPLETED SUCCESSFULLY
```

In [55]:

```
# ============================================================
# STEP 7 — REVIEW TEXT PREPROCESSING
# ============================================================

import os
import re
import pandas as pd

folder = os.path.join("output", "nlp_results")
sample_path = os.path.join(folder, "step_05_nlp_sample.csv")
reviews_df = pd.read_csv(sample_path, low_memory=False)

required_columns = [
    "Positive_Review",
    "Negative_Review",
    "Reviewer_Score"
]
```

```
missing_columns = [
    c for c in required_columns if c not in reviews_df.columns
]

if missing_columns:
    raise ValueError(f"Missing required columns: {missing_columns}")

for column in ["Positive_Review", "Negative_Review"]:
    reviews_df[column] = reviews_df[column].fillna("").astype(str)

# Remove dataset placeholders for missing comments
reviews_df["Positive_Review"] = reviews_df["Positive_Review"].replace(
    "No Positive", ""
)
reviews_df["Negative_Review"] = reviews_df["Negative_Review"].replace(
    "No Negative", ""
)

reviews_df["Review_Text"] = (
    reviews_df["Positive_Review"].str.strip()
    + " "
    + reviews_df["Negative_Review"].str.strip()
).str.strip()

def clean_review(text):
    text = str(text).lower()
    text = re.sub(r"http\S+|www\S+", " ", text)
    text = re.sub(r"[^a-z\s']", " ", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

reviews_df["Cleaned_Review"] = reviews_df["Review_Text"].apply(clean_review)

before = len(reviews_df)
reviews_df = reviews_df[
    reviews_df["Cleaned_Review"].str.len() > 0
].copy()
removed = before - len(reviews_df)

reviews_df.reset_index(drop=True, inplace=True)

output_file = os.path.join(folder, "step_07_cleaned_reviews.csv")
reviews_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 7 — REVIEW TEXT PREPROCESSING")
print("=" * 70)
print("Rows before removing empty reviews:", before)
print("Empty reviews removed:", removed)
print("Rows after preprocessing:", len(reviews_df))
print("\nReview examples:")
print(reviews_df[["Review_Text", "Cleaned_Review"]].head().to_string(index=False
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 7 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 7 — REVIEW TEXT PREPROCESSING
======================================================================
Rows before removing empty reviews: 20000
Empty reviews removed: 6
Rows after preprocessing: 19994

Review examples:

Review_Text
Cleaned_Review

good accommodation breakfast included in rate
good accommodation breakfast included in rate

Staff were vvery friendly and clean hotel
staff were vvery friendly and clean hotel

Close to the Royal Albert Hall Most uncomfortable bed I have ever slept in
close to the royal albert hall most uncomfortable bed i have ever slept in
Amazing hotel Loved all the technology and the modern feel Staff were so friendly
and helpful even offering to go and buy a bottle of wine locally as it is a halal
Room was great and the addition of a free phone to make free national and interna
tional calls as well surfing the net was a unique touch Food in the restaurant fr
om the half board menu was beautifully presented and tasty the teriyaki chicken w
as the nicest I ve had and the chocolate fondant pud oozed melted chocolate just
as it should Would totally recommend this hotel to anyone staying in that area it
s only 5 minutes walk away from Stratford High Street DLR amazing hotel loved all
the technology and the modern feel staff were so friendly and helpful even offeri
ng to go and buy a bottle of wine locally as it is a halal room was great and the
addition of a free phone to make free national and international calls as well su
rfing the net was a unique touch food in the restaurant from the half board menu
was beautifully presented and tasty the teriyaki chicken was the nicest i ve had
and the chocolate fondant pud oozed melted chocolate just as it should would tota
lly recommend this hotel to anyone staying in that area it s only minutes walk aw
ay from stratford high street dlr

Location for public transport Nice cozy room Frequency of lifts no hanging around
waiting Price of breakfast We got breakfast at one third of the price nearby When
we stayed previously breakfast was included in the price
location for public transport nice cozy room frequency of lifts no hanging around
waiting price of breakfast we got breakfast at one third of the price nearby when
we stayed previously breakfast was included in the price

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_07_c
leaned_reviews.csv

STEP 7 COMPLETED SUCCESSFULLY
```

In [56]:

```python
# ============================================================
# STEP 8 — RATING-BASED SENTIMENT LABELS
# ============================================================

import os
import pandas as pd

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(folder, "step_07_cleaned_reviews.csv")
reviews_df = pd.read_csv(input_file, low_memory=False)
```

```
reviews_df["Reviewer_Score"] = pd.to_numeric(
    reviews_df["Reviewer_Score"], errors="coerce"
)
reviews_df = reviews_df.dropna(subset=["Reviewer_Score"]).copy()

def score_to_sentiment(score):
    if score >= 7:
        return "Positive"
    elif score >= 5:
        return "Neutral"
    return "Negative"

reviews_df["Rating_Sentiment"] = (
    reviews_df["Reviewer_Score"].apply(score_to_sentiment)
)

distribution = (
    reviews_df["Rating_Sentiment"].value_counts()
    .reindex(["Positive", "Neutral", "Negative"], fill_value=0)
    .rename_axis("Sentiment").reset_index(name="Review_Count")
)

distribution["Percentage"] = (
    distribution["Review_Count"] / len(reviews_df) * 100
).round(2)

output_file = os.path.join(folder, "step_08_rating_sentiment_distribution.csv")
distribution.to_csv(output_file, index=False)

reviews_df.to_csv(input_file, index=False)

print("=" * 70)
print("STEP 8 — RATING-BASED SENTIMENT LABELS")
print("=" * 70)
print(distribution.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 8 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 8 — RATING-BASED SENTIMENT LABELS
======================================================================
Sentiment  Review_Count  Percentage
 Positive         16640       83.22
  Neutral          2498       12.49
 Negative           856        4.28

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_08_r
ating_sentiment_distribution.csv

STEP 8 COMPLETED SUCCESSFULLY
```

In [57]:

```
# ============================================================
# STEP 8.1 — INSTALL NLP DEPENDENCY
# ============================================================

%pip install nltk
```

```
Requirement already satisfied: nltk in C:\Users\admin\AppData\Local\Programs\Pyth
on\Python314\Lib\site-packages (3.10.4)
Requirement already satisfied: defusedxml in C:\Users\admin\AppData\Local\Program
s\Python\Python314\Lib\site-packages (from nltk) (0.7.1)
Requirement already satisfied: click in C:\Users\admin\AppData\Local\Programs\Pyt
hon\Python314\Lib\site-packages (from nltk) (8.5.0)
Requirement already satisfied: joblib in C:\Users\admin\AppData\Local\Programs\Py
thon\Python314\Lib\site-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in C:\Users\admin\AppData\Local\Pr
ograms\Python\Python314\Lib\site-packages (from nltk) (2026.9.29)
Requirement already satisfied: tqdm in C:\Users\admin\AppData\Local\Programs\Pyth
on\Python314\Lib\site-packages (from nltk) (4.70.1)
Requirement already satisfied: colorama in C:\Users\admin\AppData\Local\Programs
\Python\Python314\Lib\site-packages (from tqdm->nltk) (0.4.6)
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 26.2.1 -> 26.2.2
[notice] To update, run: C:\Users\admin\AppData\Local\Programs\Python\Python314\p
ython.exe -m pip install --upgrade pip
```

In [58]:

```python
# ============================================================
# STEP 9 — VADER SENTIMENT ANALYZER SETUP
# ============================================================

import os
import nltk
from datetime import datetime

folder = os.path.join("output", "nlp_results")
os.makedirs(folder, exist_ok=True)

try:
    from nltk.sentiment import SentimentIntensityAnalyzer
    nltk.download("vader_lexicon", quiet=True)

    analyzer = SentimentIntensityAnalyzer()
    setup_status = "VADER initialized successfully"
except Exception as error:
    setup_status = f"Setup failed: {error}"

setup_df = pd.DataFrame([
    {"Item": "Analyzer", "Result": "VADER"},
    {"Item": "Setup_Status", "Result": setup_status},
    {"Item": "Date", "Result": datetime.now().strftime("%Y-%m-%d %H:%M:%S")}
])

output_file = os.path.join(folder, "step_09_vader_setup.csv")
setup_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 9 — VADER SENTIMENT ANALYZER SETUP")
print("=" * 70)
print(setup_df.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))

if setup_status == "VADER initialized successfully":
    print("\nSTEP 9 COMPLETED SUCCESSFULLY")
else:
    print("\nSTEP 9 NEEDS ATTENTION")
```

```
======================================================================
STEP 9 — VADER SENTIMENT ANALYZER SETUP
======================================================================
        Item                          Result
    Analyzer                           VADER
Setup_Status VADER initialized successfully
        Date            2026-10-23 19:51:58

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_09_v
ader_setup.csv

STEP 9 COMPLETED SUCCESSFULLY
```

In [59]:

```python
# ============================================================
# STEP 10 — VADER SENTIMENT ANALYSIS
# ============================================================

import os
import pandas as pd
from nltk.sentiment import SentimentIntensityAnalyzer

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(folder, "step_07_cleaned_reviews.csv")
reviews_df = pd.read_csv(input_file, low_memory=False)

analyzer = SentimentIntensityAnalyzer()

def analyze_sentiment(text):
    compound = analyzer.polarity_scores(str(text))["compound"]

    if compound >= 0.05:
        sentiment = "Positive"
    elif compound <= -0.05:
        sentiment = "Negative"
    else:
        sentiment = "Neutral"

    return compound, sentiment

scores = reviews_df["Review_Text"].apply(analyze_sentiment)
reviews_df["VADER_Compound"] = scores.apply(lambda x: x[0])
reviews_df["VADER_Sentiment"] = scores.apply(lambda x: x[1])

output_file = os.path.join(folder, "step_10_vader_sentiment_results.csv")
reviews_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 10 — VADER SENTIMENT ANALYSIS")
print("=" * 70)
print("Reviews analyzed:", len(reviews_df))
print("\nSentiment distribution:")
print(reviews_df["VADER_Sentiment"].value_counts().to_string())
print("\nExample results:")
print(reviews_df[
    ["Review_Text", "VADER_Compound", "VADER_Sentiment"]
].head().to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 10 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 10 — VADER SENTIMENT ANALYSIS
======================================================================
Reviews analyzed: 19994

Sentiment distribution:
VADER_Sentiment
Positive    15097
Negative     2599
Neutral      2298

Example results:

Review_Text  VADER_Compound VADER_Sentiment

good accommodation breakfast included in rate          0.4404        Positive

Staff were vvery friendly and clean hotel          0.7096        Positive

Close to the Royal Albert Hall Most uncomfortable bed I have ever slept in
-0.4391        Negative
Amazing hotel Loved all the technology and the modern feel Staff were so friendly
and helpful even offering to go and buy a bottle of wine locally as it is a halal
Room was great and the addition of a free phone to make free national and interna
tional calls as well surfing the net was a unique touch Food in the restaurant fr
om the half board menu was beautifully presented and tasty the teriyaki chicken w
as the nicest I ve had and the chocolate fondant pud oozed melted chocolate just
as it should Would totally recommend this hotel to anyone staying in that area it
s only 5 minutes walk away from Stratford High Street DLR          0.9894
Positive

Location for public transport Nice cozy room Frequency of lifts no hanging around
waiting Price of breakfast We got breakfast at one third of the price nearby When
we stayed previously breakfast was included in the price          0.1531        P
ositive

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_10_v
ader_sentiment_results.csv

STEP 10 COMPLETED SUCCESSFULLY
```

In [60]:

```python
# ============================================================
# STEP 11 — SENTIMENT DISTRIBUTION SUMMARY
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

# Load sentiment analysis results
reviews_df = pd.read_csv(input_file, low_memory=False)

# Calculate sentiment counts
sentiment_order = ["Positive", "Neutral", "Negative"]
```

```
sentiment_summary = (
    reviews_df["VADER_Sentiment"]
    .value_counts()
    .reindex(sentiment_order, fill_value=0)
    .rename_axis("Sentiment")
    .reset_index(name="Review_Count")
)

# Calculate percentages
sentiment_summary["Percentage"] = (
    sentiment_summary["Review_Count"]
    / len(reviews_df) * 100
).round(2)

# Add project information
sentiment_summary["Total_Reviews"] = len(reviews_df)
sentiment_summary["Analysis_Date"] = datetime.now().strftime(
    "%Y-%m-%d %H:%M:%S"
)

# Save summary
output_file = os.path.join(
    folder, "step_11_sentiment_distribution_summary.csv"
)

sentiment_summary.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 11 — SENTIMENT DISTRIBUTION SUMMARY")
print("=" * 70)

print("\nTotal reviews analyzed:", len(reviews_df))
print("\nSentiment summary:")
print(sentiment_summary.to_string(index=False))

print("\nSaved to:")
print(os.path.abspath(output_file))

print("\nSTEP 11 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 11 — SENTIMENT DISTRIBUTION SUMMARY
======================================================================

Total reviews analyzed: 19994

Sentiment summary:
Sentiment  Review_Count  Percentage  Total_Reviews       Analysis_Date
 Positive         15097       75.51          19994 2026-10-23 19:52:10
  Neutral          2298       11.49          19994 2026-10-23 19:52:10
 Negative          2599       13.00          19994 2026-10-23 19:52:10

Saved to:
C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_11_sentiment_d
istribution_summary.csv

STEP 11 COMPLETED SUCCESSFULLY
```

In [61]:

```
# ============================================================
# STEP 12 — SENTIMENT DISTRIBUTION VISUALIZATION
```

```
# ============================================================

import os
import pandas as pd
import matplotlib.pyplot as plt

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_11_sentiment_distribution_summary.csv"
)

sentiment_summary = pd.read_csv(input_file)

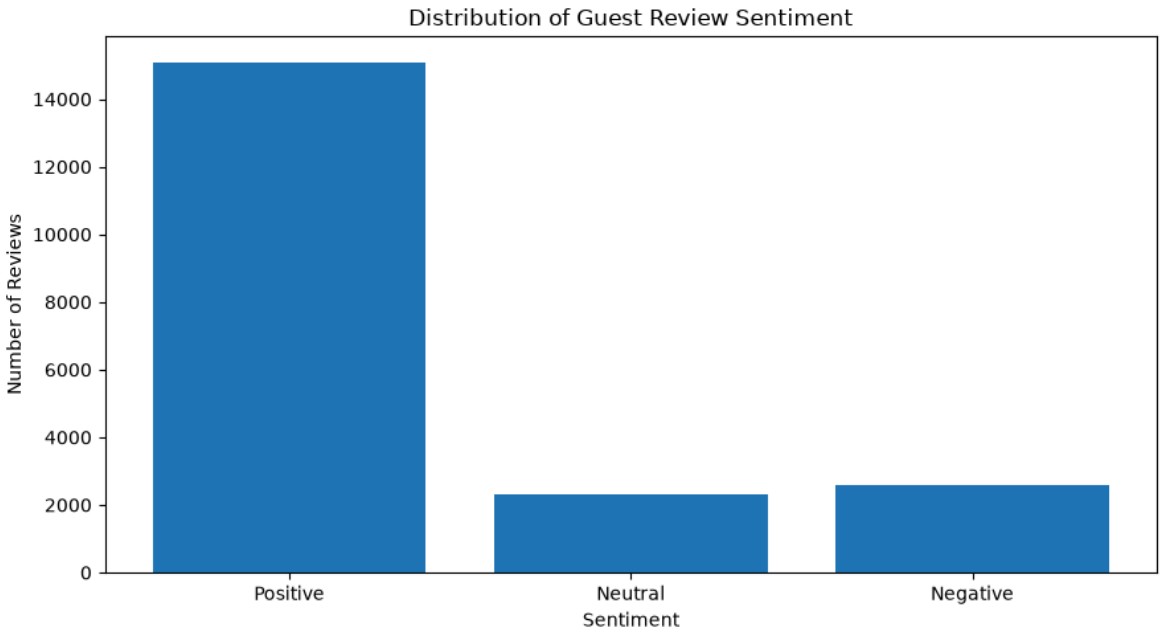
# Bar chart
plt.figure(figsize=(9, 5))
plt.bar(
    sentiment_summary["Sentiment"],
    sentiment_summary["Review_Count"]
)
plt.title("Distribution of Guest Review Sentiment")
plt.xlabel("Sentiment")
plt.ylabel("Number of Reviews")
plt.tight_layout()

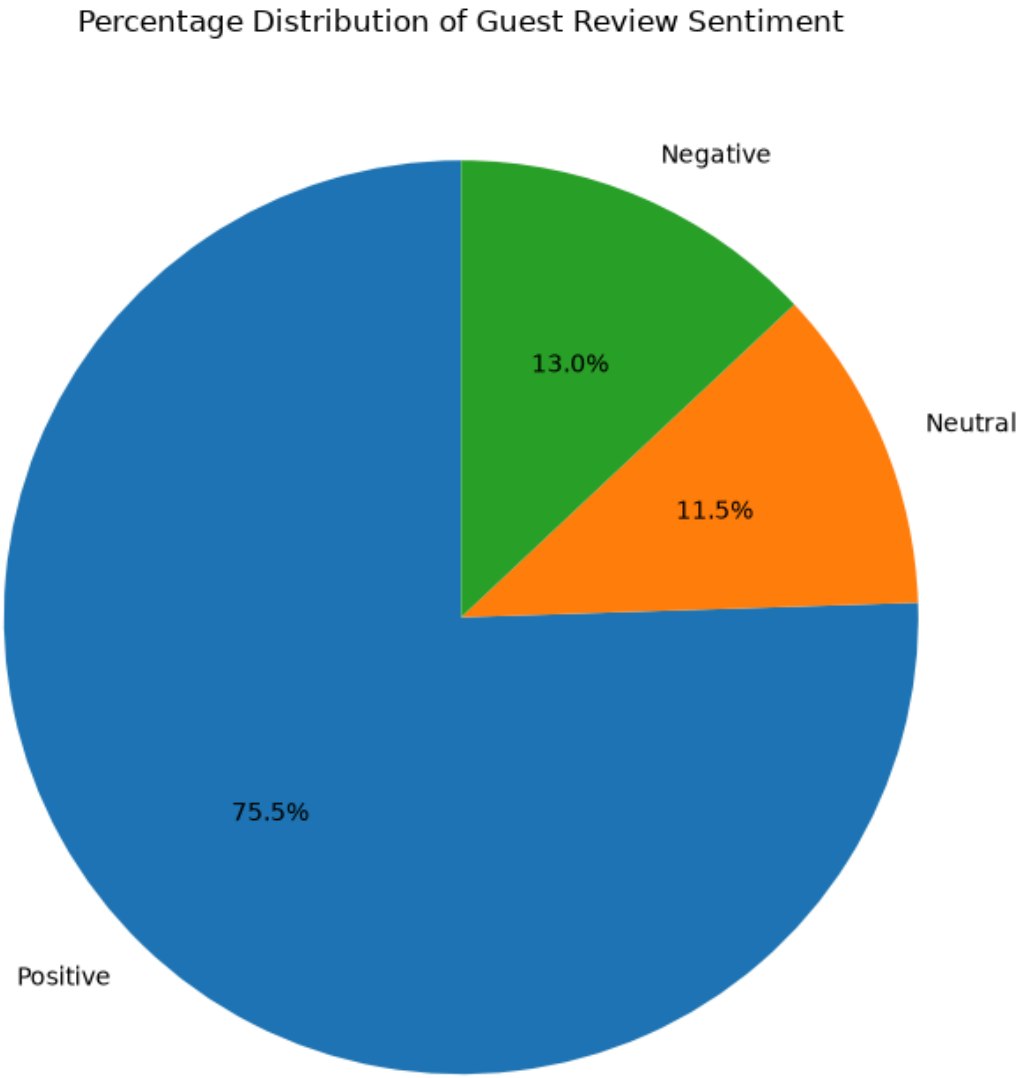
bar_file = os.path.join(
    folder, "step_12_sentiment_bar_chart.png"
)
plt.savefig(bar_file, dpi=150, bbox_inches="tight")
plt.show()

# Pie chart
plt.figure(figsize=(7, 7))
plt.pie(
    sentiment_summary["Review_Count"],
    labels=sentiment_summary["Sentiment"],
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Percentage Distribution of Guest Review Sentiment")
plt.tight_layout()

pie_file = os.path.join(
    folder, "step_12_sentiment_pie_chart.png"
)
plt.savefig(pie_file, dpi=150, bbox_inches="tight")
plt.show()

print("=" * 70)
print("STEP 12 — SENTIMENT DISTRIBUTION VISUALIZATION")
print("=" * 70)
print("Bar chart saved to:", os.path.abspath(bar_file))
print("Pie chart saved to:", os.path.abspath(pie_file))
print("\nSTEP 12 COMPLETED SUCCESSFULLY")
```


Distribution of Guest Review Sentiment
Number of Reviews
14000
12000
10000
8000
6000
4000
2000
0
Positive
Neutral
Negative
Sentiment


Percentage Distribution of Guest Review Sentiment
Negative
13.0%
Neutral
11.5%
75.5%
Positive

```
======================================================================
STEP 12 — SENTIMENT DISTRIBUTION VISUALIZATION
======================================================================
Bar chart saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results
\step_12_sentiment_bar_chart.png
Pie chart saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results
\step_12_sentiment_pie_chart.png

STEP 12 COMPLETED SUCCESSFULLY
```

In [62]:
```python
# ============================================================
# STEP 13 — RATING AND VADER SENTIMENT COMPARISON
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)

# Recreate rating-based sentiment if needed
if "Rating_Sentiment" not in reviews_df.columns:
    reviews_df["Reviewer_Score"] = pd.to_numeric(
        reviews_df["Reviewer_Score"], errors="coerce"
    )

    def score_to_sentiment(score):
        if score >= 7:
            return "Positive"
        elif score >= 5:
            return "Neutral"
        return "Negative"

    reviews_df["Rating_Sentiment"] = (
        reviews_df["Reviewer_Score"].apply(score_to_sentiment)
    )

# Compare the two methods
comparison = pd.crosstab(
    reviews_df["Rating_Sentiment"],
    reviews_df["VADER_Sentiment"],
    margins=True
)

agreement = (
    reviews_df["Rating_Sentiment"]
    == reviews_df["VADER_Sentiment"]
).mean() * 100

comparison_file = os.path.join(
    folder, "step_13_rating_vader_comparison.csv"
)

comparison.to_csv(comparison_file)
```

```
summary = pd.DataFrame([{
    "Total_Reviews": len(reviews_df),
    "Agreement_Percentage": round(agreement, 2),
    "Comparison_Date": datetime.now().strftime("%Y-%m-%d %H:%M:%S")
}])

summary_file = os.path.join(
    folder, "step_13_sentiment_agreement_summary.csv"
)
summary.to_csv(summary_file, index=False)

print("=" * 70)
print("STEP 13 — RATING AND VADER SENTIMENT COMPARISON")
print("=" * 70)
print("\nCross-tabulation:")
print(comparison.to_string())
print(f"\nAgreement percentage: {agreement:.2f}%")
print("\nComparison saved to:", os.path.abspath(comparison_file))
print("Agreement summary saved to:", os.path.abspath(summary_file))
print("\nSTEP 13 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 13 — RATING AND VADER SENTIMENT COMPARISON
======================================================================

Cross-tabulation:
VADER_Sentiment   Negative  Neutral  Positive    All
Rating_Sentiment
Negative               485      112       259    856
Neutral                870      371      1257   2498
Positive              1244     1815     13581  16640
All                   2599     2298     15097  19994

Agreement percentage: 72.21%

Comparison saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_result
s\step_13_rating_vader_comparison.csv
Agreement summary saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp
_results\step_13_sentiment_agreement_summary.csv

STEP 13 COMPLETED SUCCESSFULLY
```

In [63]:

```
# ============================================================
# STEP 14 — WORD FREQUENCY ANALYSIS
# ============================================================

import os
import re
import pandas as pd
from collections import Counter
from datetime import datetime

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)

# Check required column
```

```
if "Cleaned_Review" not in reviews_df.columns:
    raise ValueError("Cleaned_Review column not found.")

# Define common English stop words
stop_words = {
    "the", "and", "was", "for", "with", "that", "this",
    "from", "you", "are", "were", "have", "had", "has",
    "but", "not", "our", "your", "they", "their", "there",
    "very", "all", "can", "would", "could", "been", "its",
    "hotel", "room", "rooms", "stay", "stayed", "one",
    "also", "get", "got", "us", "we", "my", "me"
}

word_counter = Counter()

for text in reviews_df["Cleaned_Review"].fillna("").astype(str):
    words = re.findall(r"\b[a-z']+\b", text.lower())

    meaningful_words = [
        word for word in words
        if len(word) > 2 and word not in stop_words
    ]

    word_counter.update(meaningful_words)

top_words = word_counter.most_common(30)

word_frequency_df = pd.DataFrame(
    top_words,
    columns=["Word", "Frequency"]
)

output_file = os.path.join(
    folder, "step_14_word_frequency_analysis.csv"
)

word_frequency_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 14 — WORD FREQUENCY ANALYSIS")
print("=" * 70)

print("Reviews processed:", len(reviews_df))
print("Unique meaningful words:", len(word_counter))
print("\nTop 30 frequent words:")
print(word_frequency_df.to_string(index=False))

print("\nSaved to:")
print(os.path.abspath(output_file))

print("\nSTEP 14 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 14 — WORD FREQUENCY ANALYSIS
======================================================================
Reviews processed: 19994
Unique meaningful words: 14918

Top 30 frequent words:
       Word  Frequency
      staff       9012
   location       7807
  breakfast       5590
       good       5233
      great       4541
   friendly       3487
       nice       3162
        bed       3065
    helpful       3023
      clean       2999
comfortable       2531
  excellent       2392
      small       2382
    service       1786
    nothing       1779
     really       1772
   bathroom       1761
      which       1733
        out       1642
       when       1547
        too       1546
 everything       1488
       only       1483
        bar       1476
     lovely       1454
      close       1414
     shower       1366
    station       1346
       like       1319
      night       1304

Saved to:
C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_14_word_freque
ncy_analysis.csv

STEP 14 COMPLETED SUCCESSFULLY
```

In [64]:

```python
# ============================================================
# STEP 15 — TOP WORDS VISUALIZATION
# ============================================================

import os
import pandas as pd
import matplotlib.pyplot as plt

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_14_word_frequency_analysis.csv"
)

word_frequency_df = pd.read_csv(input_file)
```

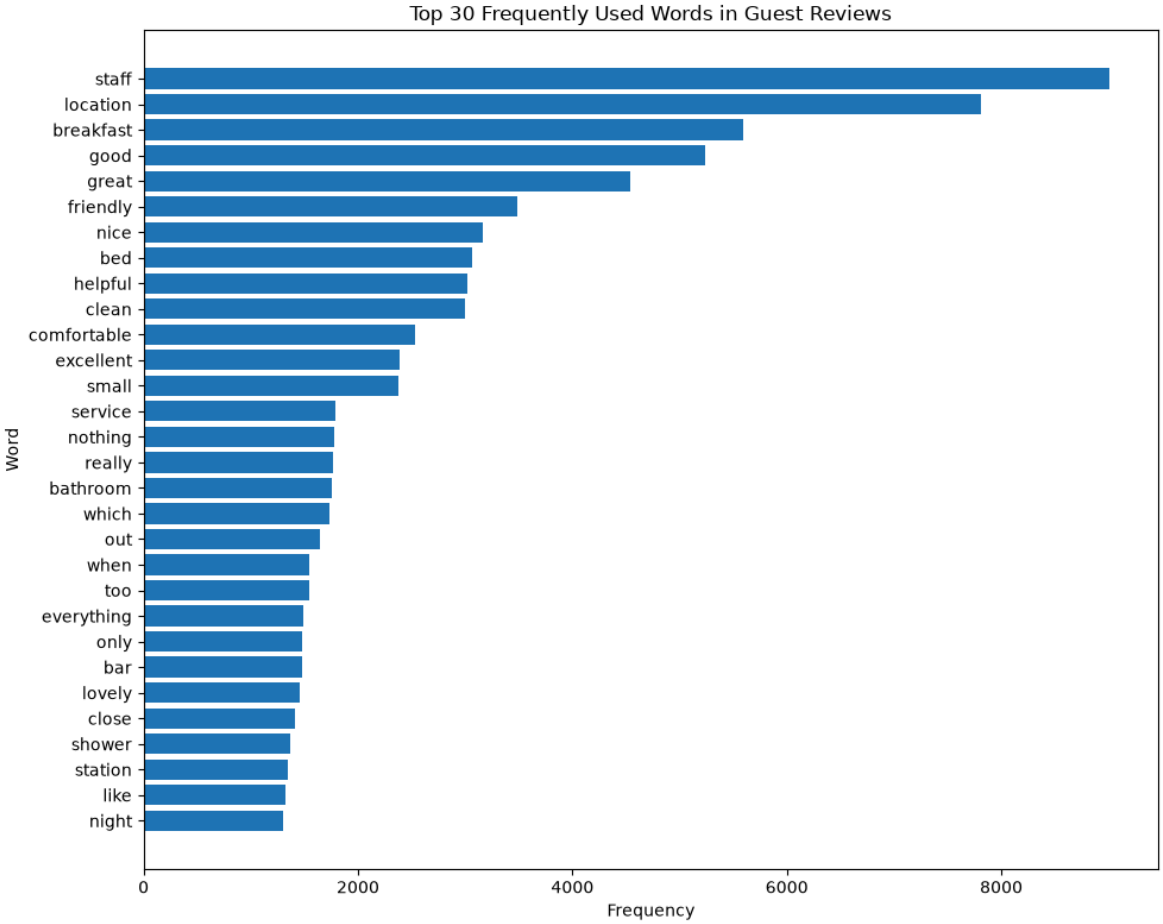
```
plot_df = word_frequency_df.sort_values(
    "Frequency", ascending=True
)

plt.figure(figsize=(10, 8))
plt.barh(plot_df["Word"], plot_df["Frequency"])
plt.title("Top 30 Frequently Used Words in Guest Reviews")
plt.xlabel("Frequency")
plt.ylabel("Word")
plt.tight_layout()

chart_file = os.path.join(
    folder, "step_15_top_words_chart.png"
)

plt.savefig(chart_file, dpi=150, bbox_inches="tight")
plt.show()

print("=" * 70)
print("STEP 15 — TOP WORDS VISUALIZATION")
print("=" * 70)
print("Words displayed:", len(word_frequency_df))
print("Chart saved to:", os.path.abspath(chart_file))
print("\nSTEP 15 COMPLETED SUCCESSFULLY")
```



```
======================================================================
STEP 15 — TOP WORDS VISUALIZATION
======================================================================
Words displayed: 30
Chart saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\ste
p_15_top_words_chart.png

STEP 15 COMPLETED SUCCESSFULLY
```

In [65]:

```python
# ============================================================
# STEP 16 — POSITIVE AND NEGATIVE FEEDBACK WORD ANALYSIS
# ============================================================

import os
import re
import pandas as pd
from collections import Counter

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)

required_columns = ["Positive_Review", "Negative_Review"]

missing = [
    col for col in required_columns
    if col not in reviews_df.columns
]

if missing:
    raise ValueError(f"Missing columns: {missing}")

stop_words = {
    "the", "and", "was", "for", "with", "that", "this",
    "from", "you", "are", "were", "have", "had", "has",
    "but", "not", "our", "your", "they", "their", "there",
    "very", "all", "can", "would", "could", "been", "its",
    "hotel", "room", "rooms", "stay", "stayed", "one",
    "also", "get", "got", "us", "we", "my", "me"
}

def count_words(series):
    counter = Counter()

    for text in series.fillna("").astype(str):
        words = re.findall(r"\b[a-z']+\b", text.lower())

        counter.update([
            word for word in words
            if len(word) > 2 and word not in stop_words
        ])

    return counter

positive_counter = count_words(reviews_df["Positive_Review"])
negative_counter = count_words(reviews_df["Negative_Review"])
```

```
positive_df = pd.DataFrame(
    positive_counter.most_common(20),
    columns=["Word", "Frequency"]
)

negative_df = pd.DataFrame(
    negative_counter.most_common(20),
    columns=["Word", "Frequency"]
)

positive_file = os.path.join(
    folder, "step_16_positive_feedback_words.csv"
)
negative_file = os.path.join(
    folder, "step_16_negative_feedback_words.csv"
)

positive_df.to_csv(positive_file, index=False)
negative_df.to_csv(negative_file, index=False)

print("=" * 70)
print("STEP 16 — POSITIVE AND NEGATIVE FEEDBACK WORD ANALYSIS")
print("=" * 70)

print("\nTop positive feedback words:")
print(positive_df.to_string(index=False))

print("\nTop negative feedback words:")
print(negative_df.to_string(index=False))

print("\nPositive words saved to:", os.path.abspath(positive_file))
print("Negative words saved to:", os.path.abspath(negative_file))
print("\nSTEP 16 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 16 — POSITIVE AND NEGATIVE FEEDBACK WORD ANALYSIS
======================================================================

Top positive feedback words:
       Word  Frequency
      staff       7450
   location       7377
       good       4434
      great       4130
   friendly       3333
  breakfast       3311
    helpful       2870
       nice       2779
      clean       2634
comfortable       2322
  excellent       2292
        bed       1852
     lovely       1360
      close       1222
    station       1184
 everything       1122
     really       1120
    perfect       1044
    service       1024
       well        936

Top negative feedback words:
     Word  Frequency
breakfast       2279
    small       1933
    staff       1562
  nothing       1518
      bed       1213
     when       1163
      too       1123
      out       1105
     only       1091
      bit       1087
 bathroom       1080
    which       1045
     didn       1030
    night        944
   little        875
     like        870
   shower        867
     good        799
     more        796
  service        762

Positive words saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_re
sults\step_16_positive_feedback_words.csv
Negative words saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_re
sults\step_16_negative_feedback_words.csv

STEP 16 COMPLETED SUCCESSFULLY
```

In [66]:
```
# ============================================================
# STEP 17 — POSITIVE VS NEGATIVE FEEDBACK VISUALIZATION
# ============================================================
```

```
import os
import pandas as pd
import matplotlib.pyplot as plt
from datetime import datetime

folder = os.path.join("output", "nlp_results")

positive_file = os.path.join(
    folder, "step_16_positive_feedback_words.csv"
)
negative_file = os.path.join(
    folder, "step_16_negative_feedback_words.csv"
)

positive_df = pd.read_csv(positive_file)
negative_df = pd.read_csv(negative_file)

# Prepare top 15 words for visualization
positive_plot = positive_df.head(15).sort_values("Frequency")
negative_plot = negative_df.head(15).sort_values("Frequency")

fig, axes = plt.subplots(1, 2, figsize=(15, 7))

axes[0].barh(
    positive_plot["Word"],
    positive_plot["Frequency"]
)
axes[0].set_title("Top Words in Positive Feedback")
axes[0].set_xlabel("Frequency")

axes[1].barh(
    negative_plot["Word"],
    negative_plot["Frequency"]
)
axes[1].set_title("Top Words in Negative Feedback")
axes[1].set_xlabel("Frequency")

plt.tight_layout()

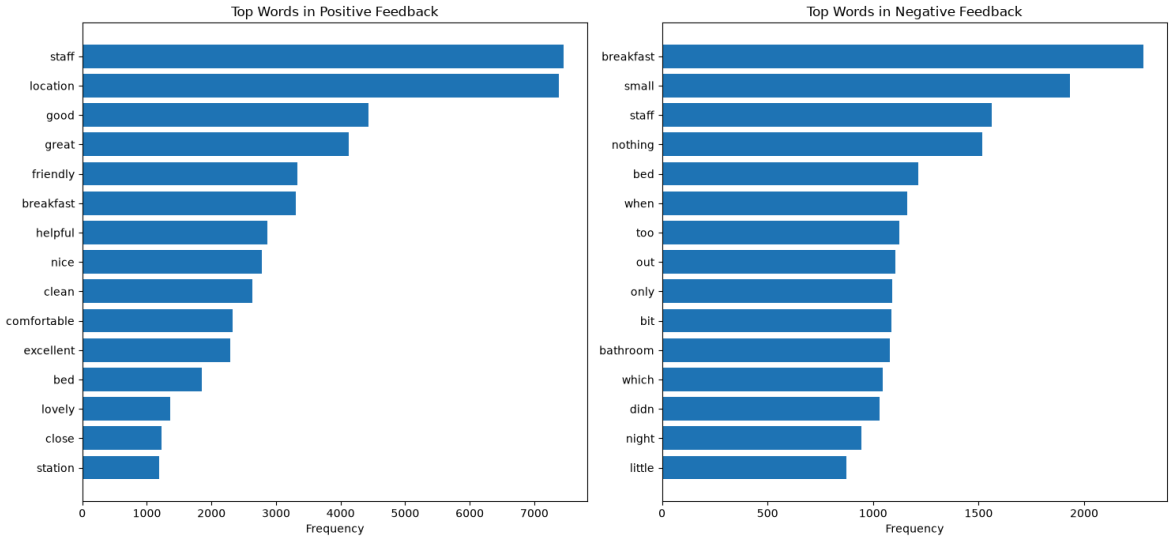
chart_file = os.path.join(
    folder, "step_17_positive_negative_comparison.png"
)
plt.savefig(chart_file, dpi=150, bbox_inches="tight")
plt.show()

summary_df = pd.DataFrame([
    {
        "Feedback_Type": "Positive",
        "Unique_Words_Among_Top_15": len(positive_plot),
        "Total_Frequency_Top_15": int(positive_plot["Frequency"].sum())
    },
    {
        "Feedback_Type": "Negative",
        "Unique_Words_Among_Top_15": len(negative_plot),
        "Total_Frequency_Top_15": int(negative_plot["Frequency"].sum())
    }
])

summary_file = os.path.join(
```

```
    folder, "step_17_feedback_comparison_summary.csv"
)
summary_df.to_csv(summary_file, index=False)

print("=" * 70)
print("STEP 17 — POSITIVE VS NEGATIVE FEEDBACK VISUALIZATION")
print("=" * 70)
print(summary_df.to_string(index=False))
print("\nChart saved to:", os.path.abspath(chart_file))
print("Summary saved to:", os.path.abspath(summary_file))
print("\nSTEP 17 COMPLETED SUCCESSFULLY")
```




```
======================================================================
STEP 17 — POSITIVE VS NEGATIVE FEEDBACK VISUALIZATION
======================================================================
Feedback_Type  Unique_Words_Among_Top_15  Total_Frequency_Top_15
     Positive                         15                   48550
     Negative                         15                   19048

Chart saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\ste
p_17_positive_negative_comparison.png
Summary saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\s
tep_17_feedback_comparison_summary.csv

STEP 17 COMPLETED SUCCESSFULLY
```

In [67]:

```
# ============================================================
# STEP 18 — SENTIMENT SCORE STATISTICS
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)

reviews_df["VADER_Compound"] = pd.to_numeric(
    reviews_df["VADER_Compound"], errors="coerce"
)
```

```
reviews_df = reviews_df.dropna(
    subset=["VADER_Compound", "VADER_Sentiment"]
)

score_summary = (
    reviews_df.groupby("VADER_Sentiment")["VADER_Compound"]
    .agg(["count", "mean", "median", "min", "max", "std"])
    .round(4)
    .reindex(["Positive", "Neutral", "Negative"])
)

score_summary.index.name = "Sentiment"
score_summary = score_summary.reset_index()

output_file = os.path.join(
    folder, "step_18_sentiment_score_statistics.csv"
)
score_summary.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 18 — SENTIMENT SCORE STATISTICS")
print("=" * 70)
print(score_summary.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 18 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 18 — SENTIMENT SCORE STATISTICS
======================================================================
Sentiment  count    mean  median     min     max    std
 Positive  15097  0.7263  0.7964  0.0507  0.9987 0.2239
  Neutral   2298  0.0002  0.0000 -0.0498  0.0498 0.0082
 Negative   2599 -0.4509 -0.4215 -0.9951 -0.0504 0.2397

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_18_sentiment_score_statistics.csv

STEP 18 COMPLETED SUCCESSFULLY
```

In [68]:

```
# ============================================================
# STEP 19 — EXAMPLE REVIEWS BY SENTIMENT
# ============================================================

import os
import pandas as pd

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)

required_columns = [
    "Review_Text", "VADER_Compound", "VADER_Sentiment"
]

missing_columns = [
    column for column in required_columns
    if column not in reviews_df.columns
```

```
]

if missing_columns:
    raise ValueError(f"Missing columns: {missing_columns}")

example_frames = []

for sentiment in ["Positive", "Neutral", "Negative"]:
    subset = reviews_df[
        reviews_df["VADER_Sentiment"] == sentiment
    ].head(5).copy()

    subset["Example_Group"] = sentiment
    example_frames.append(subset)

examples_df = pd.concat(example_frames, ignore_index=True)

examples_df = examples_df[
    ["Example_Group", "Review_Text", "VADER_Compound"]
]

output_file = os.path.join(
    folder, "step_19_example_reviews.csv"
)
examples_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 19 — EXAMPLE REVIEWS BY SENTIMENT")
print("=" * 70)

for sentiment in ["Positive", "Neutral", "Negative"]:
    print(f"\n{sentiment.upper()} REVIEW EXAMPLES")
    display(
        examples_df[examples_df["Example_Group"] == sentiment]
    )

print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 19 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 19 — EXAMPLE REVIEWS BY SENTIMENT
======================================================================

POSITIVE REVIEW EXAMPLES
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Example_Group</th> <th>Review_Text</th> <th>VADER_Compound</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

0	Positive	good accommodation breakfast included in rate	0.4404
1	Positive	Staff were vvery friendly and clean hotel	0.7096
2	Positive	Amazing hotel Loved all the technology and the...	0.9894
3	Positive	Location for public transport Nice cozy room F...	0.1531
4	Positive	Location was very good close to tube	0.4927

```
NEUTRAL REVIEW EXAMPLES
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Example_Group</th> <th>Review_Text</th> <th>VADER_Compound</th> </thead> <tbody> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> </tbody> <tbody></tbody>

Neutral	local available at short notice room small Sho...	0.0
Neutral	The staff is second to none	0.0
Neutral	A mirror at the dressing table desk would have...	0.0
Neutral	Everything N a	0.0
Neutral	Temperature control not working room is very hot	0.0

```
NEGATIVE REVIEW EXAMPLES
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Example_Group</th> <th>Review_Text</th> <th>VADER_Compound</th> </thead> <tbody> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> </tbody> <tbody></tbody>

Negative	Close to the Royal Albert Hall Most uncomforta...	-0.4391
Negative	The staff room was clean and well furnished th...	-0.0772
Negative	Central location By no means 4 standard Lots o...	-0.6486
Negative	Awful room Just by the lift which made a racke...	-0.2732
Negative	Good budget hotel Not in the prettiest neighbo...	-0.4493

```
Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_19_example_reviews.csv

STEP 19 COMPLETED SUCCESSFULLY
```

In [69]:

```
# ============================================================
# STEP 20 — NLP INTERMEDIATE OUTPUT VALIDATION
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")

required_files = [
    "step_11_sentiment_distribution_summary.csv",
    "step_12_sentiment_bar_chart.png",
    "step_12_sentiment_pie_chart.png",
    "step_13_rating_vader_comparison.csv",
    "step_13_sentiment_agreement_summary.csv",
    "step_14_word_frequency_analysis.csv",
    "step_15_top_words_chart.png",
    "step_16_positive_feedback_words.csv",
    "step_16_negative_feedback_words.csv",
    "step_17_positive_negative_comparison.png",
```

```
    "step_17_feedback_comparison_summary.csv",
    "step_18_sentiment_score_statistics.csv",
    "step_19_example_reviews.csv"
]

check_rows = []

for filename in required_files:
    full_path = os.path.join(folder, filename)

    check_rows.append({
        "File": filename,
        "Exists": os.path.exists(full_path),
        "Size_KB": (
            round(os.path.getsize(full_path) / 1024, 2)
            if os.path.exists(full_path) else None
        )
    })

check_df = pd.DataFrame(check_rows)

output_file = os.path.join(
    folder, "step_20_intermediate_output_checklist.csv"
)
check_df.to_csv(output_file, index=False)

missing_count = int((~check_df["Exists"]).sum())

print("=" * 70)
print("STEP 20 — NLP INTERMEDIATE OUTPUT VALIDATION")
print("=" * 70)
print(check_df.to_string(index=False))
print("\nRequired files:", len(required_files))
print("Missing files:", missing_count)
print("Checklist saved to:", os.path.abspath(output_file))

if missing_count == 0:
    print("\nSTEP 20 COMPLETED SUCCESSFULLY")
else:
    print("\nSTEP 20 NEEDS ATTENTION — CHECK MISSING FILES")
```

```
======================================================================
STEP 20 — NLP INTERMEDIATE OUTPUT VALIDATION
======================================================================
                                        File  Exists  Size_KB
step_11_sentiment_distribution_summary.csv    True     0.20
           step_12_sentiment_bar_chart.png    True    30.09
           step_12_sentiment_pie_chart.png    True    45.34
       step_13_rating_vader_comparison.csv    True     0.16
   step_13_sentiment_agreement_summary.csv    True     0.08
       step_14_word_frequency_analysis.csv    True     0.39
               step_15_top_words_chart.png    True    67.00
       step_16_positive_feedback_words.csv    True     0.28
       step_16_negative_feedback_words.csv    True     0.24
  step_17_positive_negative_comparison.png    True    72.05
   step_17_feedback_comparison_summary.csv    True     0.10
    step_18_sentiment_score_statistics.csv    True     0.19
               step_19_example_reviews.csv    True     2.09

Required files: 13
Missing files: 0
Checklist saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results
\step_20_intermediate_output_checklist.csv

STEP 20 COMPLETED SUCCESSFULLY
```

In [70]:

```python
# ============================================================
# STEP 21 — FREQUENT TWO-WORD PHRASE ANALYSIS
# ============================================================

import os
import re
import pandas as pd
from collections import Counter
from datetime import datetime

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)

stop_words = {
    "the", "and", "was", "for", "with", "that", "this",
    "from", "you", "are", "were", "have", "had", "has",
    "but", "not", "our", "your", "they", "their", "there",
    "very", "all", "can", "would", "could", "been", "its",
    "hotel", "room", "rooms", "stay", "stayed", "one",
    "also", "get", "got", "us", "we", "my", "me",
    "when", "which", "out", "too", "only", "little"
}

bigram_counter = Counter()

for text in reviews_df["Cleaned_Review"].fillna("").astype(str):
    words = re.findall(r"\b[a-z]+\b", text.lower())
    words = [
        word for word in words
        if len(word) > 2 and word not in stop_words
    ]
```

```
    bigram_counter.update(zip(words, words[1:]))

top_bigrams = bigram_counter.most_common(30)

bigram_df = pd.DataFrame(
    [
        {"Phrase": f"{first} {second}", "Frequency": count}
        for (first, second), count in top_bigrams
    ]
)

output_file = os.path.join(
    folder, "step_21_top_bigrams.csv"
)
bigram_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 21 — FREQUENT TWO-WORD PHRASE ANALYSIS")
print("=" * 70)
print("Reviews processed:", len(reviews_df))
print("\nTop 30 two-word phrases:")
print(bigram_df.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 21 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 21 — FREQUENT TWO-WORD PHRASE ANALYSIS
======================================================================
Reviews processed: 19994

Top 30 two-word phrases:
            Phrase  Frequency
    great location       1186
    staff friendly        989
    friendly staff        977
  friendly helpful        833
     good location        808
     helpful staff        677
     staff helpful        640
    location great        455
    breakfast good        442
excellent location        434
     location good        427
       value money        382
    good breakfast        345
   comfortable bed        333
 clean comfortable        326
  walking distance        316
   bed comfortable        314
    location staff        298
location excellent        298
  location perfect        291
  air conditioning        285
       staff great        279
  perfect location        257
  helpful friendly        255
        front desk        254
   staff excellent        253
   reception staff        250
         comfy bed        246
    location close        241
       great staff        239

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_21_top_bigrams.csv

STEP 21 COMPLETED SUCCESSFULLY
```

In [71]:

```python
# ============================================================
# STEP 22 — TOP BIGRAMS VISUALIZATION
# ============================================================

import os
import pandas as pd
import matplotlib.pyplot as plt

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(folder, "step_21_top_bigrams.csv")

bigram_df = pd.read_csv(input_file)

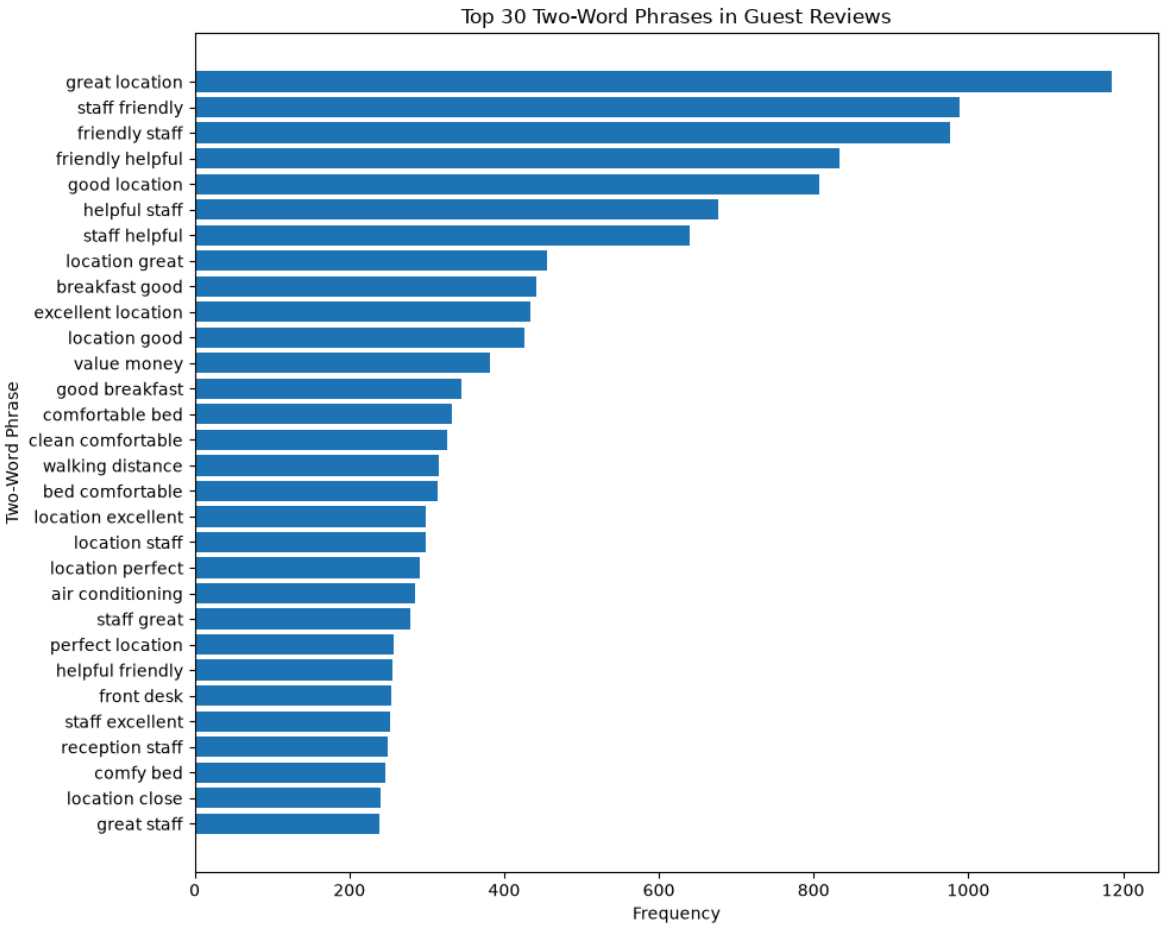
plot_df = bigram_df.sort_values("Frequency", ascending=True)

plt.figure(figsize=(10, 8))
plt.barh(plot_df["Phrase"], plot_df["Frequency"])
```

```
plt.title("Top 30 Two-Word Phrases in Guest Reviews")
plt.xlabel("Frequency")
plt.ylabel("Two-Word Phrase")
plt.tight_layout()

chart_file = os.path.join(
    folder, "step_22_top_bigrams_chart.png"
)
plt.savefig(chart_file, dpi=150, bbox_inches="tight")
plt.show()

print("=" * 70)
print("STEP 22 — TOP BIGRAMS VISUALIZATION")
print("=" * 70)
print("Phrases visualized:", len(bigram_df))
print("Chart saved to:", os.path.abspath(chart_file))
print("\nSTEP 22 COMPLETED SUCCESSFULLY")
```




```
======================================================================
STEP 22 — TOP BIGRAMS VISUALIZATION
======================================================================
Phrases visualized: 30
Chart saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\ste
p_22_top_bigrams_chart.png

STEP 22 COMPLETED SUCCESSFULLY
```

In [72]:

```
# ============================================================
# STEP 23 — GUEST FEEDBACK THEME ANALYSIS
# ============================================================

import os
```

```
import re
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)

themes = {
    "Staff": [
        "staff", "employee", "reception", "manager", "friendly"
    ],
    "Cleanliness": [
        "clean", "dirty", "cleanliness", "tidy", "filthy"
    ],
    "Location": [
        "location", "central", "station", "transport", "distance"
    ],
    "Breakfast and Food": [
        "breakfast", "food", "restaurant", "meal", "dinner"
    ],
    "Room Comfort": [
        "bed", "comfortable", "mattress", "pillow", "sleep"
    ],
    "Bathroom": [
        "bathroom", "shower", "toilet", "bath", "water"
    ],
    "Price and Value": [
        "price", "expensive", "cheap", "value", "cost"
    ],
    "Facilities": [
        "wifi", "internet", "lift", "elevator", "facilities"
    ]
}

theme_rows = []

for theme, keywords in themes.items():
    pattern = r"\b(?:" + "|".join(
        re.escape(word) for word in keywords
    ) + r")\b"

    for sentiment in ["Positive", "Neutral", "Negative"]:
        subset = reviews_df[
            reviews_df["VADER_Sentiment"] == sentiment
        ]

        matches = subset["Cleaned_Review"].fillna("").str.contains(
            pattern, regex=True, case=False
        )

        count = int(matches.sum())

        theme_rows.append({
            "Theme": theme,
            "VADER_Sentiment": sentiment,
            "Matching_Reviews": count,
```

```
            "Percentage_Within_Sentiment": round(
                count / len(subset) * 100, 2
            ) if len(subset) else 0
        })

theme_df = pd.DataFrame(theme_rows)

output_file = os.path.join(
    folder, "step_23_guest_feedback_themes.csv"
)
theme_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 23 — GUEST FEEDBACK THEME ANALYSIS")
print("=" * 70)
print(theme_df.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 23 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 23 — GUEST FEEDBACK THEME ANALYSIS
======================================================================
             Theme VADER_Sentiment  Matching_Reviews  Percentage_Within_Sentiment
             Staff        Positive              7850                        52.00
             Staff         Neutral               359                        15.62
             Staff        Negative               849                        32.67
       Cleanliness        Positive              2940                        19.47
       Cleanliness         Neutral                86                         3.74
       Cleanliness        Negative               337                        12.97
          Location        Positive              6900                        45.70
          Location         Neutral               717                        31.20
          Location        Negative               951                        36.59
Breakfast and Food        Positive              4923                        32.61
Breakfast and Food         Neutral               403                        17.54
Breakfast and Food        Negative               731                        28.13
      Room Comfort        Positive              3869                        25.63
      Room Comfort         Neutral               209                         9.09
      Room Comfort        Negative               511                        19.66
          Bathroom        Positive              2410                        15.96
          Bathroom         Neutral               207                         9.01
          Bathroom        Negative               558                        21.47
   Price and Value        Positive              1802                        11.94
   Price and Value         Neutral               170                         7.40
   Price and Value        Negative               333                        12.81
        Facilities        Positive              1598                        10.58
        Facilities         Neutral               148                         6.44
        Facilities        Negative               341                        13.12

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_23_g
uest_feedback_themes.csv

STEP 23 COMPLETED SUCCESSFULLY
```

In [73]:

```
# ============================================================
# STEP 24 — GUEST FEEDBACK THEME VISUALIZATION
# ============================================================

import os
import pandas as pd
import matplotlib.pyplot as plt
```

```
folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_23_guest_feedback_themes.csv"
)

theme_df = pd.read_csv(input_file)

pivot_df = theme_df.pivot(
    index="Theme",
    columns="VADER_Sentiment",
    values="Matching_Reviews"
).fillna(0)

for sentiment in ["Positive", "Neutral", "Negative"]:
    if sentiment not in pivot_df.columns:
        pivot_df[sentiment] = 0

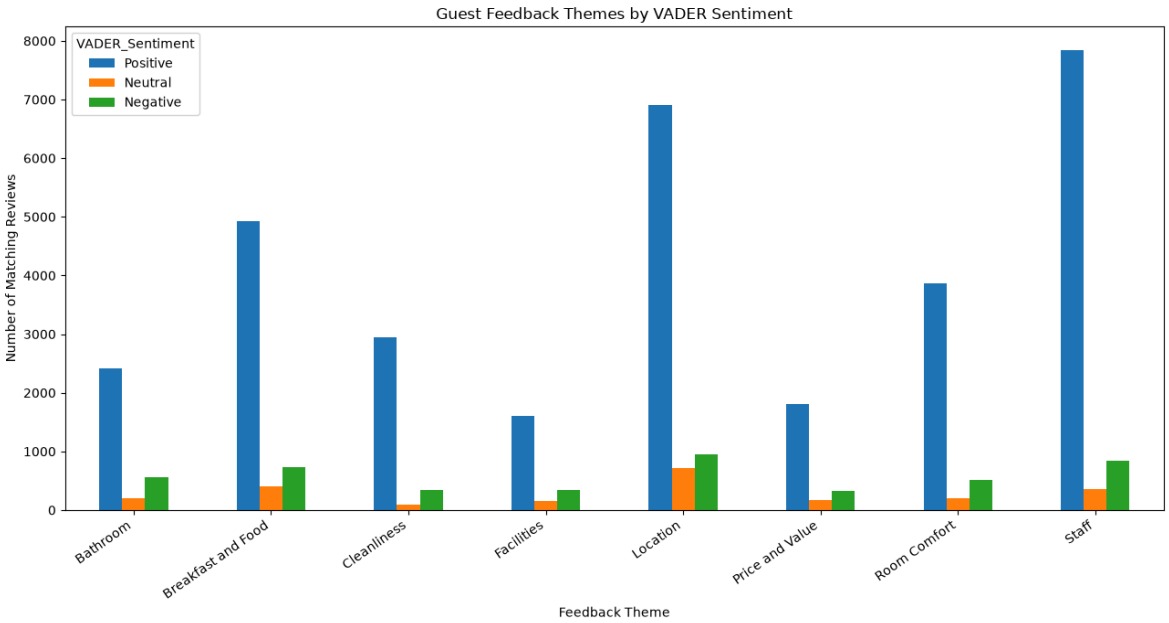
pivot_df = pivot_df[["Positive", "Neutral", "Negative"]]

ax = pivot_df.plot(
    kind="bar",
    figsize=(13, 7)
)

ax.set_title("Guest Feedback Themes by VADER Sentiment")
ax.set_xlabel("Feedback Theme")
ax.set_ylabel("Number of Matching Reviews")
plt.xticks(rotation=35, ha="right")
plt.tight_layout()

chart_file = os.path.join(
    folder, "step_24_feedback_themes_chart.png"
)
plt.savefig(chart_file, dpi=150, bbox_inches="tight")
plt.show()

print("=" * 70)
print("STEP 24 — GUEST FEEDBACK THEME VISUALIZATION")
print("=" * 70)
print("Themes analyzed:", len(pivot_df))
print("Chart saved to:", os.path.abspath(chart_file))
print("\nSTEP 24 COMPLETED SUCCESSFULLY")
```




```
======================================================================
STEP 24 — GUEST FEEDBACK THEME VISUALIZATION
======================================================================
Themes analyzed: 8
Chart saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\ste
p_24_feedback_themes_chart.png

STEP 24 COMPLETED SUCCESSFULLY
```

In [74]:

```python
# ============================================================
# STEP 25 — GUEST FEEDBACK THEME SUMMARY
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_23_guest_feedback_themes.csv"
)

theme_df = pd.read_csv(input_file)

summary_df = (
    theme_df.groupby("Theme")
    .agg(
        Total_Matching_Reviews=("Matching_Reviews", "sum"),
        Positive_Matches=("Matching_Reviews",
                          lambda x: int(x.iloc[0]))
    )
    .reset_index()
)

# Calculate positive, neutral and negative counts explicitly
pivot_df = theme_df.pivot(
    index="Theme",
    columns="VADER_Sentiment",
    values="Matching_Reviews"
).fillna(0)
```

```
for sentiment in ["Positive", "Neutral", "Negative"]:
    if sentiment not in pivot_df.columns:
        pivot_df[sentiment] = 0

pivot_df = pivot_df.reset_index()
pivot_df["Total_Matching_Reviews"] = (
    pivot_df["Positive"]
    + pivot_df["Neutral"]
    + pivot_df["Negative"]
)

pivot_df = pivot_df.sort_values(
    "Total_Matching_Reviews", ascending=False
)

output_file = os.path.join(
    folder, "step_25_theme_summary.csv"
)
pivot_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 25 — GUEST FEEDBACK THEME SUMMARY")
print("=" * 70)
print(pivot_df.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 25 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 25 — GUEST FEEDBACK THEME SUMMARY
======================================================================
             Theme  Negative  Neutral  Positive  Total_Matching_Reviews
             Staff       849      359      7850                    9058
          Location       951      717      6900                    8568
Breakfast and Food       731      403      4923                    6057
      Room Comfort       511      209      3869                    4589
       Cleanliness       337       86      2940                    3363
          Bathroom       558      207      2410                    3175
   Price and Value       333      170      1802                    2305
        Facilities       341      148      1598                    2087

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_25_t
heme_summary.csv

STEP 25 COMPLETED SUCCESSFULLY
```

In [75]:

```
# ============================================================
# STEP 26 — SEPARATE POSITIVE AND NEGATIVE COMMENT ANALYSIS
# ============================================================

import os
import pandas as pd
from nltk.sentiment import SentimentIntensityAnalyzer

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)
analyzer = SentimentIntensityAnalyzer()
```

```
required_columns = ["Positive_Review", "Negative_Review"]
missing = [c for c in required_columns if c not in reviews_df.columns]

if missing:
    raise ValueError(f"Missing required columns: {missing}")

def classify_text(text):
    text = str(text).strip()

    if not text:
        return "Empty"

    score = analyzer.polarity_scores(text)["compound"]

    if score >= 0.05:
        return "Positive"
    elif score <= -0.05:
        return "Negative"
    return "Neutral"

reviews_df["Positive_Comment_Sentiment"] = (
    reviews_df["Positive_Review"].fillna("").apply(classify_text)
)

reviews_df["Negative_Comment_Sentiment"] = (
    reviews_df["Negative_Review"].fillna("").apply(classify_text)
)

summary_rows = []

for column, comment_type in [
    ("Positive_Comment_Sentiment", "Positive comment field"),
    ("Negative_Comment_Sentiment", "Negative comment field")
]:
    counts = reviews_df[column].value_counts()

    for sentiment, count in counts.items():
        summary_rows.append({
            "Comment_Field": comment_type,
            "Detected_Sentiment": sentiment,
            "Count": int(count),
            "Percentage": round(count / len(reviews_df) * 100, 2)
        })

summary_df = pd.DataFrame(summary_rows)

output_file = os.path.join(
    folder, "step_26_separate_comment_sentiment.csv"
)
summary_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 26 — SEPARATE POSITIVE AND NEGATIVE COMMENT ANALYSIS")
print("=" * 70)
print(summary_df.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 26 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 26 — SEPARATE POSITIVE AND NEGATIVE COMMENT ANALYSIS
======================================================================
         Comment_Field Detected_Sentiment  Count  Percentage
Positive comment field           Positive  15316       76.60
Positive comment field            Neutral   2945       14.73
Positive comment field              Empty   1432        7.16
Positive comment field           Negative    301        1.51
Negative comment field           Negative   5647       28.24
Negative comment field            Neutral   5484       27.43
Negative comment field              Empty   4916       24.59
Negative comment field           Positive   3947       19.74

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_26_s
eparate_comment_sentiment.csv

STEP 26 COMPLETED SUCCESSFULLY
```

In [76]:

```python
# ============================================================
# STEP 27 — SEPARATE COMMENT SENTIMENT VISUALIZATION
# ============================================================

import os
import pandas as pd
import matplotlib.pyplot as plt

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_26_separate_comment_sentiment.csv"
)

summary_df = pd.read_csv(input_file)

pivot_df = summary_df.pivot(
    index="Detected_Sentiment",
    columns="Comment_Field",
    values="Count"
).fillna(0)

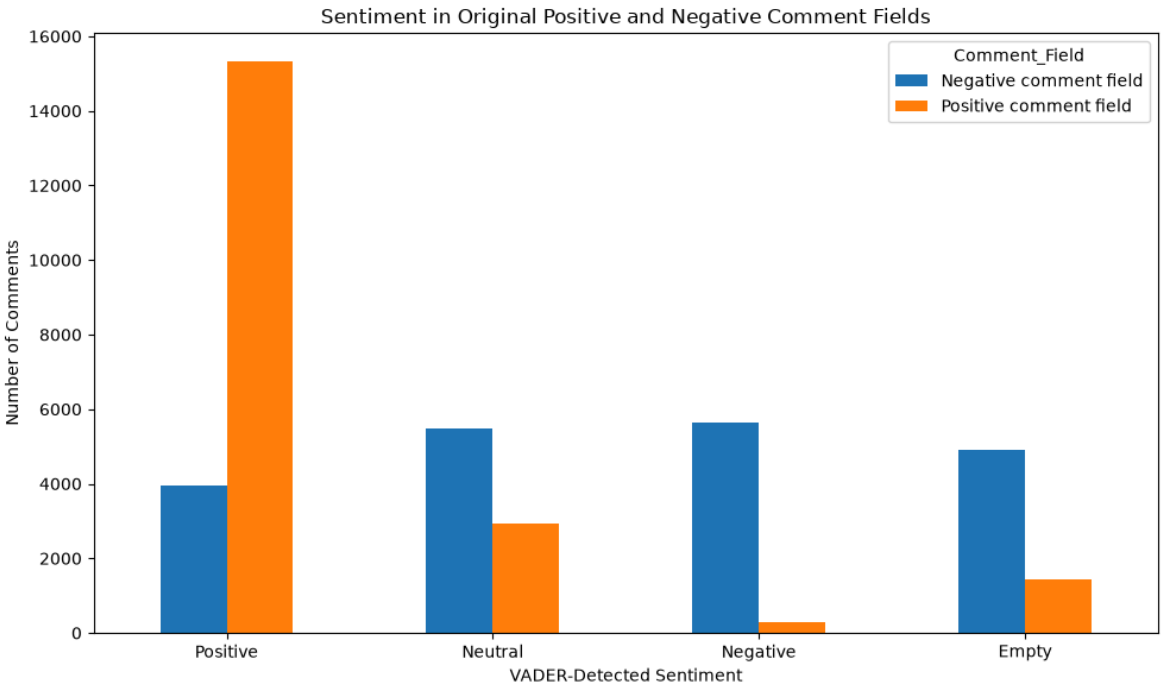
pivot_df = pivot_df.reindex(
    ["Positive", "Neutral", "Negative", "Empty"],
    fill_value=0
)

ax = pivot_df.plot(kind="bar", figsize=(10, 6))
ax.set_title("Sentiment in Original Positive and Negative Comment Fields")
ax.set_xlabel("VADER-Detected Sentiment")
ax.set_ylabel("Number of Comments")
plt.xticks(rotation=0)
plt.tight_layout()

chart_file = os.path.join(
    folder, "step_27_separate_comment_sentiment_chart.png"
)
plt.savefig(chart_file, dpi=150, bbox_inches="tight")
plt.show()

print("=" * 70)
print("STEP 27 — SEPARATE COMMENT SENTIMENT VISUALIZATION")
print("=" * 70)
```

```
print("Chart saved to:", os.path.abspath(chart_file))
print("\nSTEP 27 COMPLETED SUCCESSFULLY")
```




```
======================================================================
STEP 27 — SEPARATE COMMENT SENTIMENT VISUALIZATION
======================================================================
Chart saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\ste
p_27_separate_comment_sentiment_chart.png

STEP 27 COMPLETED SUCCESSFULLY
```

In [77]:

```
# ============================================================
# STEP 28 — REVIEWER SCORE ANALYSIS BY SENTIMENT
# ============================================================

import os
import pandas as pd
import matplotlib.pyplot as plt

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)

reviews_df["Reviewer_Score"] = pd.to_numeric(
    reviews_df["Reviewer_Score"], errors="coerce"
)

score_summary = (
    reviews_df.dropna(subset=["Reviewer_Score"])
    .groupby("VADER_Sentiment")["Reviewer_Score"]
    .agg(["count", "mean", "median", "std"])
    .round(3)
    .reindex(["Positive", "Neutral", "Negative"])
    .reset_index()
)
```

```
output_file = os.path.join(
    folder, "step_28_reviewer_score_analysis.csv"
)
score_summary.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 28 — REVIEWER SCORE ANALYSIS BY SENTIMENT")
print("=" * 70)
print(score_summary.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))

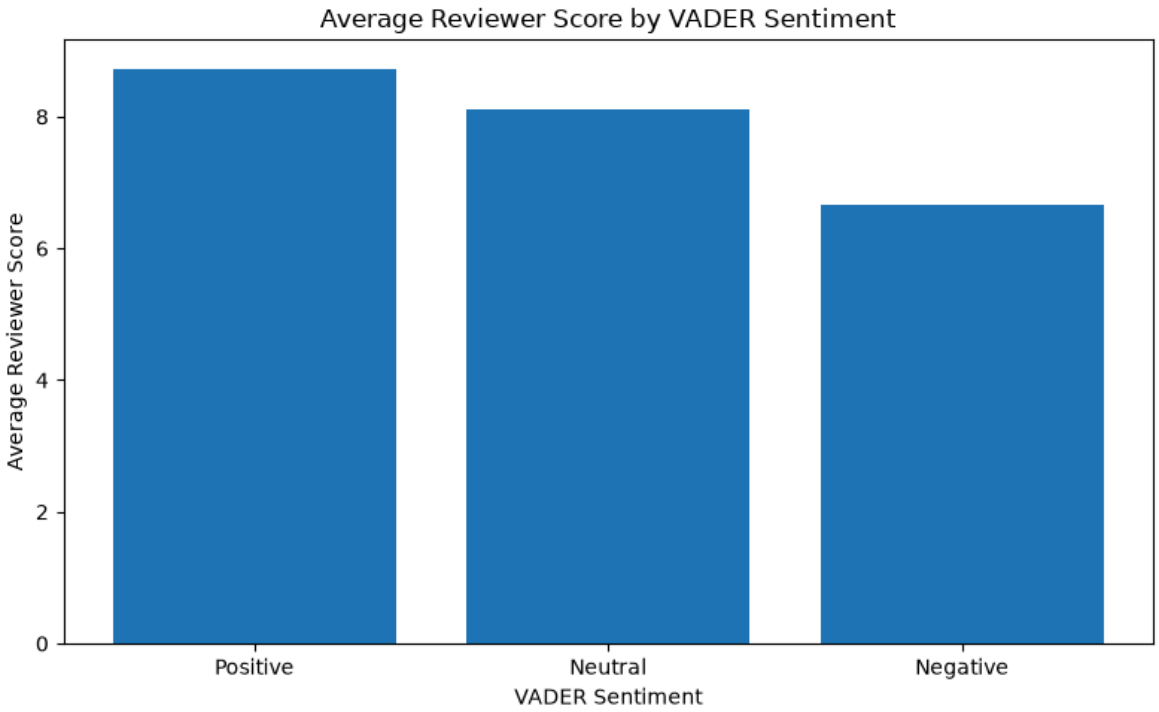
plt.figure(figsize=(8, 5))
plt.bar(
    score_summary["VADER_Sentiment"],
    score_summary["mean"]
)
plt.title("Average Reviewer Score by VADER Sentiment")
plt.xlabel("VADER Sentiment")
plt.ylabel("Average Reviewer Score")
plt.tight_layout()

chart_file = os.path.join(
    folder, "step_28_reviewer_score_chart.png"
)
plt.savefig(chart_file, dpi=150, bbox_inches="tight")
plt.show()

print("Chart saved to:", os.path.abspath(chart_file))
print("\nSTEP 28 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 28 — REVIEWER SCORE ANALYSIS BY SENTIMENT
======================================================================
VADER_Sentiment  count  mean  median   std
       Positive  15097 8.731     9.2 1.354
        Neutral   2298 8.121     8.3 1.682
       Negative   2599 6.665     6.7 1.944

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_28_r
eviewer_score_analysis.csv
```




```
Chart saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\ste
p_28_reviewer_score_chart.png

STEP 28 COMPLETED SUCCESSFULLY
```

In [78]:
```python
# ============================================================
# STEP 29 — FINAL NLP RESULTS EXPORT
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
input_file = os.path.join(
    folder, "step_10_vader_sentiment_results.csv"
)

reviews_df = pd.read_csv(input_file, low_memory=False)

# Add separate comment sentiment results if available
separate_file = os.path.join(
    folder, "step_26_separate_comment_sentiment.csv"
)

if os.path.exists(separate_file):
    from nltk.sentiment import SentimentIntensityAnalyzer
    analyzer = SentimentIntensityAnalyzer()

    def classify_text(text):
        text = str(text).strip()
        if not text:
            return "Empty"
        score = analyzer.polarity_scores(text)["compound"]
        if score >= 0.05:
            return "Positive"
        elif score <= -0.05:
            return "Negative"
```

```
        return "Neutral"

    reviews_df["Positive_Comment_Sentiment"] = (
        reviews_df["Positive_Review"].fillna("").apply(classify_text)
    )
    reviews_df["Negative_Comment_Sentiment"] = (
        reviews_df["Negative_Review"].fillna("").apply(classify_text)
    )

# Save final NLP dataset
output_file = os.path.join(
    folder, "step_29_final_nlp_results.csv"
)
reviews_df.to_csv(output_file, index=False)

export_summary = pd.DataFrame([{
    "Rows_Exported": len(reviews_df),
    "Columns_Exported": len(reviews_df.columns),
    "Output_File": os.path.basename(output_file),
    "Export_Date": datetime.now().strftime("%Y-%m-%d %H:%M:%S")
}])

summary_file = os.path.join(
    folder, "step_29_final_export_summary.csv"
)
export_summary.to_csv(summary_file, index=False)

print("=" * 70)
print("STEP 29 — FINAL NLP RESULTS EXPORT")
print("=" * 70)
print(export_summary.to_string(index=False))
print("\nDataset saved to:", os.path.abspath(output_file))
print("Export summary saved to:", os.path.abspath(summary_file))
print("\nSTEP 29 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 29 — FINAL NLP RESULTS EXPORT
======================================================================
 Rows_Exported  Columns_Exported                   Output_File         Export_Dat
e
         19994                24 step_29_final_nlp_results.csv 2026-10-23 19:52:4
5

Dataset saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\s
tep_29_final_nlp_results.csv
Export summary saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_re
sults\step_29_final_export_summary.csv

STEP 29 COMPLETED SUCCESSFULLY
```

In [79]:

```
# ============================================================
# STEP 30 — NLP FINDINGS SUMMARY
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
```

```
results_file = os.path.join(
    folder, "step_29_final_nlp_results.csv"
)
reviews_df = pd.read_csv(results_file, low_memory=False)

sentiment_counts = (
    reviews_df["VADER_Sentiment"]
    .value_counts()
    .reindex(["Positive", "Neutral", "Negative"], fill_value=0)
)

total_reviews = len(reviews_df)

findings = {
    "Project": "Hotel Booking Cancellation Prediction and Analysis",
    "Stage": "Activity 2 – NLP Guest Feedback Analysis",
    "Dataset": "Hotel_Reviews.csv",
    "Reviews_Analyzed": total_reviews,
    "Positive_Reviews": int(sentiment_counts["Positive"]),
    "Neutral_Reviews": int(sentiment_counts["Neutral"]),
    "Negative_Reviews": int(sentiment_counts["Negative"]),
    "Positive_Percentage": round(
        sentiment_counts["Positive"] / total_reviews * 100, 2
    ) if total_reviews else 0,
    "Neutral_Percentage": round(
        sentiment_counts["Neutral"] / total_reviews * 100, 2
    ) if total_reviews else 0,
    "Negative_Percentage": round(
        sentiment_counts["Negative"] / total_reviews * 100, 2
    ) if total_reviews else 0,
    "Rating_VADER_Agreement_Percentage": 72.21,
    "NLP_Method": "VADER lexicon-based sentiment analysis",
    "Theme_Method": "Keyword-based exploratory theme analysis",
    "Limitation": (
        "Sample-based analysis; sentiment labels are automated; "
        "review dataset is not linked to booking cancellation records"
    ),
    "Date": datetime.now().strftime("%Y-%m-%d %H:%M:%S")
}

findings_df = pd.DataFrame(
    list(findings.items()),
    columns=["Item", "Result"]
)

output_file = os.path.join(
    folder, "step_30_nlp_findings_summary.csv"
)
findings_df.to_csv(output_file, index=False)

print("=" * 70)
print("STEP 30 – NLP FINDINGS SUMMARY")
print("=" * 70)
print(findings_df.to_string(index=False))
print("\nSaved to:", os.path.abspath(output_file))
print("\nSTEP 30 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 30 — NLP FINDINGS SUMMARY
======================================================================
                            Item
Result
                         Project
Hotel Booking Cancellation Prediction and Analysis
                           Stage
Activity 2 — NLP Guest Feedback Analysis
                         Dataset
Hotel_Reviews.csv
               Reviews_Analyzed
19994
               Positive_Reviews
15097
                Neutral_Reviews
2298
               Negative_Reviews
2599
            Positive_Percentage
75.51
             Neutral_Percentage
11.49
            Negative_Percentage
13.0
Rating_VADER_Agreement_Percentage
72.21
                     NLP_Method
VADER lexicon-based sentiment analysis
                    Theme_Method
Keyword-based exploratory theme analysis
                      Limitation Sample-based analysis; sentiment labels are aut
omated; review dataset is not linked to booking cancellation records
                           Date
2026-10-23 19:52:45

Saved to: C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_30_n
lp_findings_summary.csv

STEP 30 COMPLETED SUCCESSFULLY
```

In [80]:

```python
# ============================================================
# STEP 31 — FINAL NLP OUTPUT FILE VALIDATION
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
os.makedirs(folder, exist_ok=True)

required_files = [
    "step_01_nlp_project_setup.csv",
    "step_10_vader_sentiment_results.csv",
    "step_11_sentiment_distribution_summary.csv",
    "step_12_sentiment_bar_chart.png",
    "step_12_sentiment_pie_chart.png",
    "step_13_rating_vader_comparison.csv",
    "step_13_sentiment_agreement_summary.csv",
```

```
    "step_14_word_frequency_analysis.csv",
    "step_15_top_words_chart.png",
    "step_16_positive_feedback_words.csv",
    "step_16_negative_feedback_words.csv",
    "step_17_positive_negative_comparison.png",
    "step_17_feedback_comparison_summary.csv",
    "step_18_sentiment_score_statistics.csv",
    "step_19_example_reviews.csv",
    "step_20_intermediate_output_checklist.csv",
    "step_21_top_bigrams.csv",
    "step_22_top_bigrams_chart.png",
    "step_23_guest_feedback_themes.csv",
    "step_24_feedback_themes_chart.png",
    "step_25_theme_summary.csv",
    "step_26_separate_comment_sentiment.csv",
    "step_27_separate_comment_sentiment_chart.png",
    "step_28_reviewer_score_analysis.csv",
    "step_28_reviewer_score_chart.png",
    "step_29_final_nlp_results.csv",
    "step_29_final_export_summary.csv",
    "step_30_nlp_findings_summary.csv"
]

rows = []

for filename in required_files:
    path = os.path.join(folder, filename)
    exists = os.path.isfile(path)

    rows.append({
        "File": filename,
        "Exists": exists,
        "Size_KB": round(os.path.getsize(path) / 1024, 2)
        if exists else None
    })

check_df = pd.DataFrame(rows)

output_file = os.path.join(
    folder, "step_31_final_nlp_output_checklist.csv"
)
check_df.to_csv(output_file, index=False)

missing_count = int((~check_df["Exists"]).sum())

print("=" * 70)
print("STEP 31 — FINAL NLP OUTPUT FILE VALIDATION")
print("=" * 70)
print(check_df.to_string(index=False))
print("\nTotal files checked:", len(required_files))
print("Files found:", int(check_df["Exists"].sum()))
print("Missing files:", missing_count)
print("\nChecklist saved to:")
print(os.path.abspath(output_file))

if missing_count == 0:
    print("\nSTEP 31 COMPLETED SUCCESSFULLY")
else:
    print("\nSTEP 31 NEEDS ATTENTION — CHECK MISSING FILES")
```

```
======================================================================
STEP 31 — FINAL NLP OUTPUT FILE VALIDATION
======================================================================
                                          File  Exists  Size_KB
                step_01_nlp_project_setup.csv    True     0.40
          step_10_vader_sentiment_results.csv    True 16637.01
  step_11_sentiment_distribution_summary.csv    True     0.20
              step_12_sentiment_bar_chart.png    True    30.09
              step_12_sentiment_pie_chart.png    True    45.34
          step_13_rating_vader_comparison.csv    True     0.16
     step_13_sentiment_agreement_summary.csv    True     0.08
          step_14_word_frequency_analysis.csv    True     0.39
                  step_15_top_words_chart.png    True    67.00
          step_16_positive_feedback_words.csv    True     0.28
          step_16_negative_feedback_words.csv    True     0.24
    step_17_positive_negative_comparison.png    True    72.05
     step_17_feedback_comparison_summary.csv    True     0.10
      step_18_sentiment_score_statistics.csv    True     0.19
                  step_19_example_reviews.csv    True     2.09
   step_20_intermediate_output_checklist.csv    True     0.62
                     step_21_top_bigrams.csv    True     0.61
                step_22_top_bigrams_chart.png    True   100.10
           step_23_guest_feedback_themes.csv    True     0.80
           step_24_feedback_themes_chart.png    True    69.20
                    step_25_theme_summary.csv    True     0.29
      step_26_separate_comment_sentiment.csv    True     0.38
step_27_separate_comment_sentiment_chart.png    True    47.89
          step_28_reviewer_score_analysis.csv    True     0.13
             step_28_reviewer_score_chart.png    True    30.10
                step_29_final_nlp_results.csv    True 16961.69
             step_29_final_export_summary.csv    True     0.11
             step_30_nlp_findings_summary.csv    True     0.61

Total files checked: 28
Files found: 28
Missing files: 0

Checklist saved to:
C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_31_final_nlp_o
utput_checklist.csv

STEP 31 COMPLETED SUCCESSFULLY
```

In [81]:

```
# ============================================================
# STEP 32 — FINAL NLP ANALYSIS REPORT
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")

findings_file = os.path.join(
    folder, "step_30_nlp_findings_summary.csv"
)
themes_file = os.path.join(
    folder, "step_25_theme_summary.csv"
)
scores_file = os.path.join(
```

```
    folder, "step_28_reviewer_score_analysis.csv"
)

findings_df = pd.read_csv(findings_file)
themes_df = pd.read_csv(themes_file)
scores_df = pd.read_csv(scores_file)

findings = dict(zip(findings_df["Item"], findings_df["Result"]))

def get_value(key, default="Not available"):
    return findings.get(key, default)

# Rank themes by number of matching reviews
themes_ranked = themes_df.sort_values(
    "Total_Matching_Reviews", ascending=False
)

report_lines = [
    "=" * 75,
    "FINAL NLP GUEST FEEDBACK ANALYSIS REPORT",
    "=" * 75,
    "",
    "Project: Hotel Booking Cancellation Prediction and Analysis",
    "Stage: Activity 2 — NLP Guest Feedback Analysis",
    f"Report generated: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}",
    "",
    "1. OBJECTIVE",
    "Analyze hotel guest review text using VADER sentiment analysis,",
    "word frequencies, phrase frequencies, and keyword-based themes.",
    "",
    "2. DATASET SUMMARY",
    f"Reviews analyzed: {get_value('Reviews_Analyzed')}",
    f"Dataset: {get_value('Dataset')}",
    "",
    "3. SENTIMENT DISTRIBUTION",
    f"Positive reviews: {get_value('Positive_Reviews')} "
    f"({get_value('Positive_Percentage')}%)",
    f"Neutral reviews: {get_value('Neutral_Reviews')} "
    f"({get_value('Neutral_Percentage')}%)",
    f"Negative reviews: {get_value('Negative_Reviews')} "
    f"({get_value('Negative_Percentage')}%)",
    "",
    "4. RATING AND SENTIMENT COMPARISON",
    "Rating-VADER agreement: "
    f"{get_value('Rating_VADER_Agreement_Percentage')}%",
    "",
    "5. REVIEWER SCORES BY SENTIMENT",
    scores_df.to_string(index=False),
    "",
    "6. MOST FREQUENT FEEDBACK THEMES",
    themes_ranked.head(8).to_string(index=False),
    "",
    "7. MAIN INTERPRETATION",
    "Most reviews were classified as positive by VADER in this sample.",
    "Staff, location, breakfast, room comfort, and cleanliness were",
    "among the explored feedback themes. Theme counts may overlap.",
    "",
    "8. LIMITATIONS",
    "VADER is a lexicon-based sentiment tool and can misinterpret context,",
    "sarcasm, mixed opinions, and short comments.",
```

```
    "Rating-based labels are proxies, not independently annotated truth.",
    "Themes were identified using predefined keywords, not a trained",
    "topic model or manual validation.",
    "The review dataset is separate from the booking cancellation dataset;",
    "this analysis does not directly measure cancellation behavior.",
    "",
    "9. CONCLUSION",
    "NLP provided a structured overview of guest sentiment and recurring",
    "feedback topics. Results should be interpreted as exploratory findings.",
    "",
    "=" * 75
]

report_text = "\n".join(report_lines)

output_file = os.path.join(
    folder, "step_32_final_nlp_analysis_report.txt"
)

with open(output_file, "w", encoding="utf-8") as file:
    file.write(report_text)

print(report_text)
print("\nReport saved to:")
print(os.path.abspath(output_file))
print("\nSTEP 32 COMPLETED SUCCESSFULLY")
```

```
================================================================================
FINAL NLP GUEST FEEDBACK ANALYSIS REPORT
================================================================================

Project: Hotel Booking Cancellation Prediction and Analysis
Stage: Activity 2 – NLP Guest Feedback Analysis
Report generated: 2026-10-23 19:52:45

1. OBJECTIVE
Analyze hotel guest review text using VADER sentiment analysis,
word frequencies, phrase frequencies, and keyword-based themes.

2. DATASET SUMMARY
Reviews analyzed: 19994
Dataset: Hotel_Reviews.csv

3. SENTIMENT DISTRIBUTION
Positive reviews: 15097 (75.51%)
Neutral reviews: 2298 (11.49%)
Negative reviews: 2599 (13.0%)

4. RATING AND SENTIMENT COMPARISON
Rating–VADER agreement: 72.21%

5. REVIEWER SCORES BY SENTIMENT
VADER_Sentiment  count  mean  median   std
       Positive  15097 8.731     9.2 1.354
        Neutral   2298 8.121     8.3 1.682
       Negative   2599 6.665     6.7 1.944

6. MOST FREQUENT FEEDBACK THEMES
             Theme  Negative  Neutral  Positive  Total_Matching_Reviews
             Staff       849      359      7850                    9058
          Location       951      717      6900                    8568
Breakfast and Food       731      403      4923                    6057
      Room Comfort       511      209      3869                    4589
       Cleanliness       337       86      2940                    3363
          Bathroom       558      207      2410                    3175
   Price and Value       333      170      1802                    2305
        Facilities       341      148      1598                    2087

7. MAIN INTERPRETATION
Most reviews were classified as positive by VADER in this sample.
Staff, location, breakfast, room comfort, and cleanliness were
among the explored feedback themes. Theme counts may overlap.

8. LIMITATIONS
VADER is a lexicon-based sentiment tool and can misinterpret context,
sarcasm, mixed opinions, and short comments.
Rating-based labels are proxies, not independently annotated truth.
Themes were identified using predefined keywords, not a trained
topic model or manual validation.
The review dataset is separate from the booking cancellation dataset;
this analysis does not directly measure cancellation behavior.

9. CONCLUSION
NLP provided a structured overview of guest sentiment and recurring
feedback topics. Results should be interpreted as exploratory findings.

================================================================================
```

```
Report saved to:
C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_32_final_nlp_analysis_report.txt

STEP 32 COMPLETED SUCCESSFULLY
```

In [82]:

```python
# ============================================================
# STEP 33 — FINAL NLP COMPLETION SUMMARY
# ============================================================

import os
import pandas as pd
from datetime import datetime

folder = os.path.join("output", "nlp_results")
os.makedirs(folder, exist_ok=True)

summary = {
    "Project": "Hotel Booking Cancellation Prediction and Analysis",
    "Stage": "Activity 2 — NLP Guest Feedback Analysis",
    "Dataset": "Hotel_Reviews.csv",
    "Reviews_Analyzed": 19994,
    "Positive_Percentage": 75.51,
    "Neutral_Percentage": 11.49,
    "Negative_Percentage": 13.00,
    "Rating_VADER_Agreement_Percentage": 72.21,
    "Sentiment_Analysis_Completed": "Yes",
    "Word_Frequency_Analysis_Completed": "Yes",
    "Bigram_Analysis_Completed": "Yes",
    "Feedback_Theme_Analysis_Completed": "Yes",
    "Visualizations_Completed": "Yes",
    "Final_Report_Created": "Yes",
    "Output_Checklist": "All 28 required files present",
    "Output_Files_Missing": 0,
    "Method": "VADER lexicon-based sentiment analysis",
    "Theme_Method": "Keyword-based exploratory analysis",
    "Limitation": (
        "Review dataset is separate from booking data; "
        "NLP findings do not directly predict cancellations"
    ),
    "Status": (
        "NLP implementation and reporting completed; "
        "final review recommended"
    ),
    "Date": datetime.now().strftime("%Y-%m-%d %H:%M:%S")
}

summary_df = pd.DataFrame(
    list(summary.items()),
    columns=["Item", "Result"]
)

output_file = os.path.join(
    folder, "step_33_final_nlp_completion_summary.csv"
)

summary_df.to_csv(output_file, index=False)

print("=" * 70)
```

```
print("STEP 33 — FINAL NLP COMPLETION SUMMARY")
print("=" * 70)
print(summary_df.to_string(index=False))
print("\nSaved to:")
print(os.path.abspath(output_file))
print("\nSTEP 33 COMPLETED SUCCESSFULLY")
```

```
======================================================================
STEP 33 — FINAL NLP COMPLETION SUMMARY
======================================================================
                            Item
Result
                         Project                                                H
otel Booking Cancellation Prediction and Analysis
                           Stage
Activity 2 — NLP Guest Feedback Analysis
                         Dataset
Hotel_Reviews.csv
                Reviews_Analyzed
19994
             Positive_Percentage
75.51
              Neutral_Percentage
11.49
             Negative_Percentage
13.0
Rating_VADER_Agreement_Percentage
72.21
     Sentiment_Analysis_Completed
Yes
Word_Frequency_Analysis_Completed
Yes
        Bigram_Analysis_Completed
Yes
Feedback_Theme_Analysis_Completed
Yes
         Visualizations_Completed
Yes
             Final_Report_Created
Yes
                Output_Checklist
All 28 required files present
             Output_Files_Missing
0
                          Method
VADER lexicon-based sentiment analysis
                     Theme_Method
Keyword-based exploratory analysis
                       Limitation Review dataset is separate from booking data; N
LP findings do not directly predict cancellations
                          Status                              NLP implementation
and reporting completed; final review recommended
                            Date
2026-10-23 19:52:45

Saved to:
C:\Users\admin\OneDrive\Desktop\Activity-2\output\nlp_results\step_33_final_nlp_c
ompletion_summary.csv

STEP 33 COMPLETED SUCCESSFULLY
```

In [ ]: